



Meta-analysis

The relationship between Impulse Control Disorders and cognitive dysfunctions in Parkinson's Disease: A meta-analysis[☆]



Gabriella Santangelo^{a,*}, Simona Raimo^a, Paolo Barone^b

^a Department of Psychology, University of Campania Luigi Vanvitelli, Caserta, Italy

^b Department of Medicine, Center for Neurodegenerative Diseases (CEMAND), University of Salerno, Italy

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ABSTRACT

Impulse Control Disorders (ICD) are associated with impairment in cognitive flexibility and cortical inhibition. In Parkinson's Disease (PD) the relationship between ICD and cognitive dysfunctions is still unclear: some studies found different cognitive profiles between Parkinsonians with and without ICD, whereas others did not. Moreover, findings from studies on ICD in PD are conflicting on which cognitive function is altered. A meta-analysis of 34 studies was performed to shed light on relationship between ICD and cognitive dysfunctions and to reveal the cognitive function compromised in Parkinsonians with ICD. Data were analysed in global cognitive functioning, memory, executive functions, attention/working memory, language, and visuospatial functions. Significant relationship between ICD and dysfunction of abstraction ability/concept formation, set-shifting, visuospatial/constructional abilities and decision-making was found. These findings suggested that people affected by PD with specific frontal dysfunctions are more vulnerable to develop ICD when they take antiparkinsonian drug. Evaluation of specific cognitive functions in routine clinical practice might help to detect those people with PD susceptible to ICD before treating them with antiparkinsonian drugs.

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* Corresponding author at: Department of Psychology, University of Campania Luigi Vanvitelli, Caserta, Viale Ellittico 31, 81100 Caserta, Italy.
E-mail address: gabriella.santangelo@unina2.it (G. Santangelo).

1. Introduction

In Parkinson's Disease (PD) Impulse Control Disorders (ICD) include pathological gambling (PG), hypersexuality (HS), compulsive shopping or buying (CS), compulsive or binge eating (CE).

These disturbances can have devastating psychosocial consequences for the person involved and her/his family. The ICD are characterized by impulsive aspects because of their strong relationship between high novelty-seeking and impulsivity traits and also by compulsive aspects because of their similarities to substance abuse (i.e. experience of withdrawal symptoms and/or the development of tolerance) (Leeman and Potenza, 2012).

The ICD in PD are considered as effect side of dopaminergic treatment (Voon et al., 2011a); in particular, a stimulation of DA in a relatively intact reward system may lead a hyper-dopaminergic state, a condition predisposing to development of ICD.

As regard factors associated with ICD in PD, decision-making and cognitive control processes were found dysfunctional in people with PD and with ICD. While findings of impaired decision-making in PD are consistent across studies (Voon et al., 2011a; Voon et al., 2010; Djamshidian et al., 2010, 2012; Housden et al., 2010; Leroi et al., 2013; Rossi et al., 2010), reports of cognitive dysfunctions associated with ICD in PD are conflicting (Santangelo et al., 2013a, 2013b). Some studies revealed that patients with ICD performed significantly worse than patients without ICD on several frontal tasks (Santangelo et al., 2009; Vitale et al., 2011; Voon et al., 2010; Djamshidian et al., 2010; Biundo et al., 2011, 2015; Tessitore et al., 2016; Yoo et al., 2015a), whereas some studies failed to reveal a significant association between ICD and cognitive dysfunctions or revealed that patients with ICD functioned cognitively better than those without ICDs (Siri et al., 2010, 2015; Pineau et al., 2016). Therefore, until now the role of cognitive dysfunction as contributor to ICD is unclear and should be better elucidated.

Another issue, which remains unclear and deserves further studies, is the type of cognitive functions altered in Parkinsonians with ICD. About this, studies reporting an association between ICD and cognitive dysfunctions revealed dysfunction of inhibition (Vitale et al., 2011; Yoo et al., 2015a), generativity (Santangelo et al., 2009; Tessitore et al., 2016), set-shifting (Vitale et al., 2011; Biundo et al., 2011, 2015; Santangelo et al., 2009; Tessitore et al., 2016; Voon et al., 2010), spatial planning (Vitale et al., 2011), abstraction ability/concept formation (Vitale et al., 2011; Santangelo et al., 2009; Tessitore et al., 2016), reasoning (Santangelo et al., 2009; Tessitore et al., 2016) and memory (Djamshidian et al., 2010; Tessitore et al., 2016; Vitale et al., 2011; Santangelo et al., 2009; Voon et al., 2010). This inconsistency among the studies may depend on a host of factors associated with cognitive impairment including, for example, selection criteria and characteristics of patients, comorbid disorders, and sample size.

A meta-analytic review aims to shed light on the relationship between ICDs and cognitive impairment in people with PD and on which cognitive domain is damaged in this type of Parkinsonians. Moreover, a meta-analysis of these relationships may help clarify findings where statistical power has been inadequate in individual studies, where findings are not consistent across. We examined the influence of demographic and clinical confounders (i.e. clinical stage, treatment, age at onset of PD and disease duration) on cognitive performances in people with PD.

1.1. Objectives

Aims of meta-analysis were: 1. to systematically investigate the relationship between ICD and cognition in PD; 2. to identify alterations of specific cognitive functions associated with ICDs (CE, HS, PG, punding, compulsive shopping or buying) in PD.

2. Method

2.1. Search strategy

A systematic literature search was performed on 19 June 2016 using PsychInfo (PROQUEST), PubMed, Scopus restricted to papers in English from peer-reviewed journals, supplemented by hand searches of reference lists from included and seminal papers. We excluded conference proceedings, theses, and case studies. Where the same data was presented in more than one publication, we used the primary (first) publication.

Search terms are shown in supplemental material 1 and produced a total of 1112 articles. After exclusion of duplicates and reports judged to be irrelevant based on title and abstract screening, 42 articles were retrieved for detailed inspection (Fig. 1). Two reviewers independently evaluated all papers in according to inclusion and exclusion criteria summarized below. All aspects of study selection, extraction, and assessment were performed by two reviewers working independently (GS, SR). Disagreements between reviewers were resolved through discussion or with recourse to a third arbitrator if required (PB).

2.2. Study eligibility criteria

Studies were included if: 1. Participants were adults diagnosed with idiopathic PD; 2. The study was based on a neuropsychological perspective or included a cognitive evaluation; 3. Cognition were measured by validated neuropsychological tests (i.e. Mini Mental State Examination, MMSE); 4. Samples were independent (for prospective study we used baseline data); 5. Cognitive comparisons between Parkinsonians with and without ICDs were reported statistically. The studies were also excluded if: 1. The study included atypical PD or parkinsonian syndromes; 2. The study investigated only the neuronal or neurochemical basis of ICD in PD; 3. The study concentrated on cognitive functions without linking them directly to ICD in PD; 4. The study investigated cognition by self-report assessment tools.

2.3. Outcomes

For each study, the primary outcome was neuropsychological test scores. Over thirty individual neuropsychological variables were found to be employed in all studies included in the present meta-analysis. We grouped individual cognitive task under six cognitive domains and their subdomains (Table 1) based on approach reported in Litvan et al. (2012) or on indication provided in primary study, or when no literature was available, on agreement between two authors (GS, PB) with specific expertise in neuropsychological assessment in PD. Subsequently, we analysed 1. Global Cognitive Function; 2. General Executive Functioning; 3. the sub-domains of memory (i.e. verbal and spatial short term and long-term recall) 4. the sub-domains of executive functions (i.e. abstraction ability/concept formation, shifting, updating, inhibition, generativity and fluid reasoning), 5. Language, 6. Processing Speed/Complex Attention/Working Memory; 7. Visuospatial/constructional abilities; 8. Decision Making.

When in a primary study the same outcome was evaluated using more than one instrument, extracted data were reported from the most relevant instrument, which was determined by consensus of two authors (GS, PB).

2.4. Data extraction and coding

Data extracted and coded from the final articles included: 1. characteristics of the publication: (i.e. authors, publication status, year of publication, journal); 2. characteristics of the sample (i.e.

Table 1

Cognitive domains examined in this review, neuropsychological tests used in included the primary studies.

Cognitive Outcomes	Test	Primary studies
Global Cognitive Function	MMSE	Antonini et al. (2011); Bentivoglio et al. (2013); Biundo et al. (2011); Biundo et al. (2015); Cera et al. (2014); Cerasa et al. (2014); Cilia et al. (2008); Claassen et al. (2015); Housden et al. (2010); Leroi et al. (2011); Mack et al. (2013); Pellicano et al. (2015); Pettorruso et al. (2014); Piray et al. (2014); Politis et al. (2013); Pontieri et al. (2015); Rossi et al. (2010); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Tessitore et al. (2016); Vitale et al. (2011); Voon et al. (2010); Voon et al. (2007); Yoo et al. (2015a); Yoo et al. (2015b)
	MoCA	Voon et al. (2011b)
	ACE-R	Hurt et al. (2014)
Memory		
Verbal Short Term Memory	Digit Span- Forward	Bentivoglio et al. (2013); Biundo et al. (2011); Biundo et al. (2015); Cerasa et al. (2014); Siri et al. (2010); Siri et al. (2015); Piray et al. (2014); Djamshidian et al. (2010); Yoo et al. (2015a)
	Verbal Span	Santangelo et al. (2009)
Spatial Short Term Memory	Test di Corsi	Bentivoglio et al. (2013); Biundo et al. (2011); Biundo et al. (2015); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Tessitore et al. (2016)
Verbal Long Term Memory	15 RAVLT- verbal recall	Bentivoglio et al. (2013); Biundo et al. (2011); Cerasa et al. (2014); Pontieri et al. (2015); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Tessitore et al. (2016); Vitale et al. (2011)
	Hopkins-verbal recall	Mack et al. (2013)
	Memory prose test	Biundo et al. (2015)
	SVLT-verbal recall	Yoo et al. (2015a)
Spatial Long Term Memory	ROCF-delayed recall	Biundo et al. (2011); Biundo et al. (2015); Pontieri et al. (2015); Santangelo et al. (2009); Yoo et al. (2015a)
Executive Functions		
General	FAB	Antonini et al. (2011); Bentivoglio et al. (2013); Biundo et al. (2011); Cerasa et al. (2014); Cilia et al. (2008); Pettorruso et al. (2014); Rossi et al. (2010); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Voon et al. (2007)
Shifting	TMT: B	Biundo et al. (2011); Biundo et al. (2015); Mack et al. (2013); Tessitore et al. (2016)
	Intra-and extradimensional set-shifting	Voon et al. (2010)
	TMT:B-A	Biundo et al. (2011); Biundo et al. (2015); Mack et al. (2013); Leroi et al. (2013); Santangelo et al. (2009); Vitale et al. (2011); Pineau et al. (2016)
Abstraction Ability*	WCST-categories	Santangelo et al. (2009); Tessitore et al. (2016); Vitale et al. (2011)
	MCST-categories	Bentivoglio et al. (2013); Leroi et al. (2013); Pontieri et al. (2015); Rossi et al. (2010); Pineau et al. (2016)
Updating	Digit Span-Backwards	Bentivoglio et al. (2013); Biundo et al. (2011); Cerasa et al. (2014); Djamshidian et al. (2010); Piray et al. (2014); Siri et al. (2015); Yoo et al. (2015a)
Inhibition	Stroop Test	Djamshidian et al. (2011); Yoo et al. (2015a); Vitale et al. (2011); Cera et al. (2014); Pontieri et al. (2015); Rossi et al. (2010); Tessitore et al. (2016); Biundo et al. (2011); Biundo et al. (2015)
	Short Stroop Test	Bentivoglio et al. (2013)
	Connor's Performance Test	Pineau et al. (2016)
Generativity	Stop Signal Task-motor	Claassen et al. (2015); Leroi et al. (2013)
	Phonological fluency	Bentivoglio et al. (2013); Biundo et al. (2011); Biundo et al. (2015); Cerasa et al. (2014); Leroi et al. (2011); Leroi et al. (2013); Mack et al. (2013); Pontieri et al. (2015); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Tessitore et al. (2016); Yoo et al. (2015a); Pineau et al. (2016)
	Semantic fluency	Bentivoglio et al. (2013); Biundo et al. (2011); Biundo et al. (2015); Mack et al. (2013); Pontieri et al. (2015); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Tessitore et al. (2016); Yoo et al. (2015a)
- Fluid Reasoning	Raven's Progressive Matrices	Bentivoglio et al. (2013); Biundo et al. (2011); Cilia et al. (2008); Santangelo et al. (2009); Siri et al. (2010); Siri et al. (2015); Tessitore et al. (2016)
Processing Speed/ Complex Attention/Working Memory		
	TMT: part A	Biundo et al. (2011); Biundo et al. (2015); Mack et al. (2013)
	Attentive matrices	Siri et al. (2010); Tessitore et al. (2016); Vitale et al. (2011)
	Digit Span Forward	Bentivoglio et al. (2013); Cerasa et al. (2014); Djamshidian et al. (2010); Piray et al. (2014); Siri et al. (2015); Yoo et al. (2015a)
	Spatial working memory	Voon et al. (2010)
	Digit n-back	Leroi et al. (2013); Leroi et al. (2011)
	Working memory test	Djamshidian et al. (2012)
	Conner's Performance Test	Pineau et al. (2016)
Visuospatial Ability		
	Judgment Line Orientation	Cerasa et al. (2014)
	ROCF-copy task	Biundo et al. (2011); Biundo et al. (2015); Pontieri et al. (2015); Santangelo et al. (2009); Tessitore et al. (2016); Vitale et al. (2011); Yoo et al. (2015a)
	Constructional Apraxia test	Bentivoglio et al. (2013)
Language		
	Boston Naming test	Yoo et al. (2015a)
	Oral verbal naming	Bentivoglio et al. (2013)
	Token test	Cerasa et al. (2014)
	Similarities	Biundo et al. (2011)
	Naming	Biundo et al. (2015); Siri et al. (2010)
Decision making (positive direction: ICD+ are more impulsive than ICD-)		
	Kirby Delayed Discounting questionnaire	Housden et al. (2010)
	Delayed Discounting Test	Leroi et al. (2013); Voon et al. (2010); Cera et al. (2014)
	Iowa Gambling Test	Rossi et al. (2010); Bentivoglio et al. (2013); Pineau et al. (2016)
	Beads Test	Djamshidian et al. (2012)
	Risk task	Voon et al. (2011a); Djamshidian et al. (2010)

*Abstraction Ability/Concept Formation); MMSE (Mini Mental State Examination); MoCA (Montreal Cognitive Assessment (ACE-R (Addenbrooke's Cognitive Exam-Revised); RAVLT (Rey's Auditory Verbal Learning Test); SVLT (Souel Verbal Learning Test); ROCF (Rey–Osterrieth complex figure test); FAB (Frontal Assessment Battery); TMT:B (Trail Making Test: part B); TMT:B-A (Trail Making Test: part B-part A); WCST (Wisconsin Card Sorting Test); MCST (Modified Card Sorting Test).

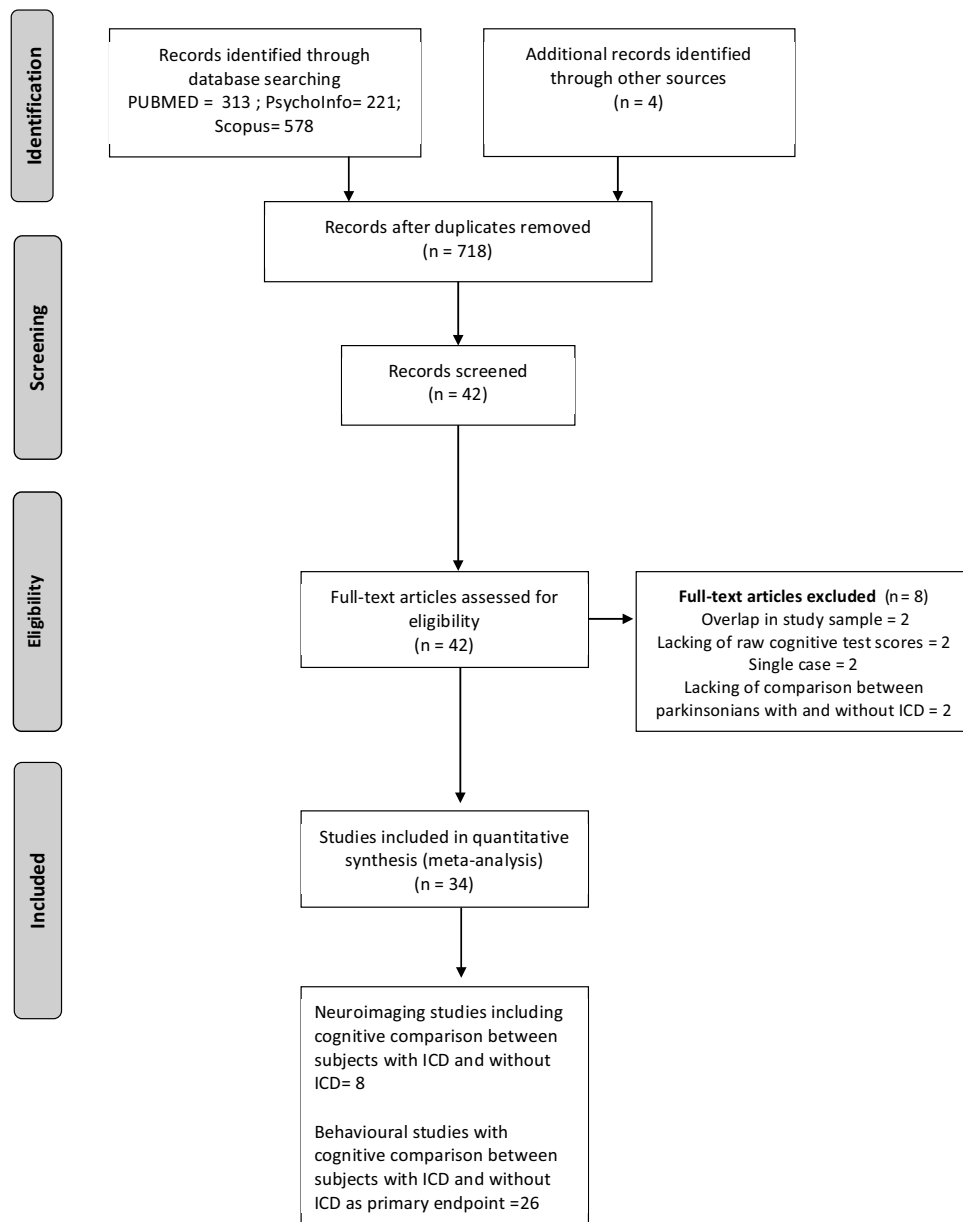


Fig. 1. Flow diagram based on PRISMA statement (www.prisma-statement.org).

total sample size, gender was coded as the frequency of men in a sample; age at evaluation, age at PD onset, disease duration, severity of motor symptoms (evaluated by motor part of Unified Parkinson's Disease Rating Scale, UPDRS) and stage of PD (evaluated by Hoehn & Yahr stage) was coded as the mean and standard deviation); 3. dementia used as exclusion criterion in primary studies (used or not used in the primary studies) and the type of the diagnostic criteria used to identify dementia in each primary study. 4. tools used to identify people with and without ICDs; 5. neuropsychological tests (i.e. cognitive tests).

2.5. Statistical analyses

We synthesized study data using meta-analytic methods. Initially, we computed the effect sizes from data reported in the articles (e.g., means and standard deviations; *p* values) using Hedges' *g* unbiased approach (similar to the Cohen *d* statistic). Negative values of the Hedges' *g* indicated that people with ICDs perform worse than people without ICDs. The conventions used to

interpret Hedges' *g* are similar to Cohen's *d*. According to Cohen's (Cohen, 1988) criteria, values <0.20 are considered small effects, values of about 0.50 moderate effects, and values of about 0.80 large effects. For each effect size, 95% confidence interval, variance, standard error and statistical significance were computed. Effect sizes were pooled across studies for obtaining an overall effect size with the inverse-variance method. The random-effects model was used as a conservative approach to account for different sources of variation among studies and to generalize the meta-analytic finding beyond the studies included here.

To assess heterogeneity among studies, *Q* and *I*² statistics were computed (Huedo-Medina et al., 2006). A significant *Q* value indicates a lack of homogeneity of findings among studies; the proportion of observed variance that reflects real differences in effect sizes was estimated by *I*². A value of 25, 50, and 75% was considered as low, moderate and high, respectively (Higgins et al., 2003). To further explain heterogeneity across study findings, we conducted moderator analysis with 12 continuous moderators (i.e., age at evaluation, gender, education, age at onset of PD, disease dura-

tion, severity of motor symptoms assessed by Unified Parkinson's Disease Rating Scale (UPDRS), stage of disease assessed by Hoehn & Yahr (H&Y), antiparkinsonian treatments: Levodopa dosage mg/d, Total Levodopa Equivalent Daily dose, Dopamine agonist daily dose) which were assessed by meta-regressions. We conducted sensitivity analyses to check the stability of study findings, computing how the overall effect size would change removing one study at a time. Finally, publication bias analysis was performed in order to control for the fact that published studies may have a larger mean effect size than unpublished studies (Rothstein et al., 2005). Two methods were applied: the funnel plot and trim and fill procedure. The first is a scatter plot of the effects sizes estimated from individual studies against a measure of their precision (e.g., their standard errors). To evaluate the funnel plot more reliably, we employed the Egger's regression method (Egger et al., 1997), which to statistically test the asymmetry of the funnel plot, with nonsignificant results indicative of absence of publication bias. The trim and fill procedure is an iterative non-parametric statistical technique which evaluates the effect of potential data censoring on the result of the meta-analyses (Duval and Tweedie, 2000). In this method, the absence of publication bias is indicated by zero trimmed studies, or in the presence of trimmed studies, by trivial difference between the observed and the estimated effect sizes (Duval, 2005). Statistical analyses were conducted with the meta-analytic software ProMeta 3.0.

3. Results

3.1. Literature search

The Fig. 1 reported the flow diagram based on PRISMA statement. The initial search identified 718 possible articles after the removal of duplicates. 676 articles were excluded after abstract selection. After full-text selection, 42 studies were considered eligible but we excluded 8 articles because of overlap in the study sample (2 articles), lacking of raw cognitive test scores (2 articles), single case (2 articles) and lacking of a cognitive comparison between Parkinsonians with and without ICD (2 articles). Out of 36 primary studies, 26 studies investigated the association between ICDs and cognitive performance as primary endpoint, whereas in 8 neuroimaging studies this issue was a secondary endpoint (Cerasa et al., 2014; Cilia et al., 2008; Pellicano et al., 2015; Biundo et al., 2015, 2011; Tessitore et al., 2016; Yoo et al., 2015a, 2015b). However, we decided to include them in the analysis because a cognitive comparison between ICD and non-ICD people was reported. Moreover, three of the 36 studies (Pettorruso et al., 2014; Pontieri et al., 2015; Vitale et al., 2011) examined cognitive profile of people with ICDs, dividing them into two groups (PG and ICD-NOS group, Pettorruso et al., 2014; Pontieri et al., 2015) and into PG, HS, CE and multiple ICDs groups (Vitale et al., 2011). As the type of ICD was not relevant to this review, groups were collapsed into ICD+ group by calculating the pooled means and standard deviations from statistics provided in each paper. In other two studies the mean and standard deviation of reaction time on Stroop test (Djamshidian et al., 2011), working memory test and beads test (Djamshidian et al., 2012) obtained by ICD+ and <ICD- groups (in on medication) were not reported, but we estimated results from graphics. In a study (Djamshidian et al., 2010), data from Working Memory Test were reported in the form of z-scores; to include that study in the meta-analysis, we decided to enter *p*-value and sample size of both ICD+ and ICD- groups. In three studies (Hurt et al., 2014; Yoo et al., 2015a; Pineau et al., 2016) the data from demographic, clinical and cognitive variables were reported in median and range; we estimated mean and standard deviation following specific formulas (Hozo et al., 2005). Finally, 34 studies were included in the meta-analysis. The main characteristics of selected studies are shown in Table 2.

3.1.1. Descriptive characteristics of studies included in the meta-analysis

The present meta-analysis is based on data extracted from 34 studies published between 2007–2016, which matched the eligibility criteria. Sample sizes of 2807 people with PD, with means ages comprised between 51.52 and 70.37 years, with mean ages of PD onset comprised between 43.9–61 years, with mean UPDRS- motor score comprised between 11.7–36.7, with mean H&Y stages comprised between 1.3–2.4 (in 13 studies the data of H&Y stage were not reported). Moreover, dementia was considered to be an exclusion criterion in 30 primary studies (83%), while 3 studies (8.33%) did not exclude demented patients and 1 study (2.7%) did not report this issue. Out of 30 primary studies excluding demented subjects, 11 studies employed clinical criteria for diagnosis of dementia, 12 studies employed global cognitive battery (11 studies used MMSE and 1 used Addenbrooke's Cognitive Examination-Revised version), 4 studies used both clinical diagnostic criteria and MMSE. Instead, 3 studies did not specify the tools used to identify dementia.

3.2. Meta-analytic results

Taking into account cognitive outcomes, in Table 1 we showed the reference of the primary studies that investigated a cognitive outcome and we described the cognitive tests used in each primary study in order to evaluate a specific cognitive outcome. We conducted 18 main meta-analyses examining the association between ICDs and several cognitive defects. Detailed results of a total of 18 meta-analyses are reported in Table 3; results regarding differences in cognitive performance between ICD+ and ICD- groups are displayed in Figs. 2–16.

In detail, ICD+ group were found to be moderately but significantly more impaired in tasks involving abstraction ability/concept formation (Fig. 2), updating (Fig. 3), set-shifting (evaluated by means TMT-B and TMT:B-A, Fig. 4) and visuospatial/constructional abilities (Fig. 5). The Heterogeneity was between low and moderate for Abstraction ability/concept formation, and Visuospatial/Constructional abilities, high for Set-shifting evaluated by TMT:B-A; however, Egger's test was not significant (Table 3). After removing the study by Leroi et al. (2013) from the meta-analysis, we found a slight increase in effect size (Effect size: -0.62 ; 95% Confidence Intervals Lower Limit: -0.98 , Upper Limit: -0.26 ; $p=0.001$). Heterogeneity became slightly smaller ($I^2=57\%$) and Egger's test remained not significant ($t=0.21$; $p=0.846$).

We did not find a significant difference on Global and Frontal Cognitive Functioning (Figs. 6 and 7) Language (Fig. 8), Processing speed/Attention/Working Memory (Fig. 9), Spatial and Verbal Long- and Short-Term memory (Figs. 10 and 11) and even on remaining Executive Functions (i.e. Fluid Reasoning, Inhibition, Generativity, Figs. 12–15). The Heterogeneity was between low and moderate. Egger's test was not significant for all, but Fluid Reasoning. For the latter outcome, there were no trimmed studies indicating no publication bias.

A separate meta-analysis conducted for the Trail Making Test (Fig. 4): part B showed significant difference between ICD+ and ICD-. Heterogeneity was low, and Egger's test was not significant.

As for decision-making (Fig. 16), ICD+ group were found to be moderately but significantly more impulsive in making a choice; this finding was affected by moderate heterogeneity alone. There was no publication bias. After removing the study by Rossi et al. (2010) from the meta-analysis, we found a slight reduction in effect size (Effect Size: 0.39 ; 95% Confidence Intervals Lower Limit: 0.039 , Upper Limit: 0.76 ; $p=0.035$). Heterogeneity became slightly smaller ($I^2=60\%$) and Egger's test remained not significant ($t=-0.05$; $p=0.965$).

Table 2
Characteristics of primary studies included in the meta-analysis.

Reference	Participants; (Gender: n Men)	mean age, years (SD)	mean age at PD onset, years (SD)	Mean PD duration (SD)	Mean Education (SD)	Mean UPDRS motor score	H&Y	Antiparkinsonian treatment	Tool assessing ICDs	Cognitive tasks	Dementia as exclusion criterion/tools used to identify dementia
Antonini et al. (2011)	18 ICD+ (15)	58.3 (9.7)	NR	19.3 (16.5)	13.4 (4.7)	16.5 (8.1)	1.4 (0.5)	No treatment	MIDI, SOGS, diagnostic criteria	MMSE, FAB	Yes/DSM IV criteria
Bentivoglio et al. (2013)	85 ICD– (52)	60.9 (9.1)		14.6 (15)	11.1 (4.5)	16.4 (8.9)	1.6 (0.5)				
	17 ICD+ (14)	62.0 (10.1)	NR	6.9 (3.8)	8.7 (3.7)	23.8 (11)	2.0 (0.8)	Total LEDD mg: 606.1 (319.2) Levodopa dosage mg/d: 539 (264.3) DA agonist dosage: 172.9 (112.2)	diagnostic criteria	RAVLT, Immediate visual memory, Digit and Corsi span forward, and backward, RCPM, verb and noun naming, MWCST, Stroop test, Double barrage, IGT, FAB, Constructional, orofacial and ideomotor apraxia, Phon- and semantic verbal fluency,	Yes/MMSE
	17 ICD– (11)	63.94 (9.2)		7.3 (4.4)	10.17 (4.4)	22.5 (6.9)	2.3 (0.5)	Total LEDD mg: 616.2 (367.8) Levodopa dosage mg/d: 455.7 (299.0) DA agonist dosage: 192.5 (88.5)			
Biundo et al. (2011)	33 ICD+ (18)	61.27 (10.19)	53.18 (10.56)	8.81 (4.79)	11.78 (3.91)	30.23 (13.21)	NR	Total LEDD mg: 556.79 (304.63) DA agonist dosage: 186.51 (149.34)	MIDI, diagnostic criteria	MMSE, RCPM, RAVL Phon- and semantic verbal fluency, Stroop test, TMT, FAB, ROCF, similarities, Digit span forward and backward, Corsi's test,	Yes/DSM IV criteria and MMSE
	24 ICD– (17)	70.37 (6.80)	60.54 (9.98)	8.87 (5.43)	10.37 (4.8)	32.28 (12.83)		Total LEDD mg: 497.36 (341.18) DA agonist dosage: 165.83 (108.78)			
Biundo et al. (2015)	58 ICD+ (38)	60.3 (9.3)	50.1 (12.1)	9 (5.5)	10.9 (4.3)	26.7 (16.5)	2.4 (0.7)	Total LEDD mg: 923.1 (474.1) DA agonist dosage: 163.7 (111.3)	MDI, QUIP-RS, diagnostic criteria	TMT, Digit Span, Digit Ordering Test, Corsi's test, Stroop, ROCF, Phon- and semantic verbal fluency, Prose memory, CDT, WPAT, Naming, VOSP letters	Yes/diagnostic criteria for PDD (Dubois et al., 2007)
	52 ICD– (32)	63.1 (10.2)	54.7 (11.6)	8 (5.7)	11.3 (4.7)	28.5 (12.3)	2.3 (0.7)	Total LEDD mg: 722.6 (498.5) DA agonist dosage: 148.9 (105.0)			
Cera et al. (2014)	10 PG+ alone (7)	60.6 (6.8)	NR	28.2 (12.3)	11.7 (2.6)	20.5 (6.8)	1.9 (0.2)	Total LEDD mg: 294.5 (123.1)	SOGS, diagnostic criteria, QUIP-RS	Monetary reward/loss risk-taking task; Stroop Test, Emotional Stroop Test (all computer versions)	Yes/MMSE
	9 ICD+ PG- (6)	59.3 (6.8)		29 (8.5)	10.3 (3.2)	21.4 (4.2)	1.7 (0.3)	Total LEDD mg: 283.3 (132.9)			
	14 ICD– (7)	59.0 (9.5)		27.2 (8.4)	11.7 (1.9)	21.6 (6.9)	1.7 (0)	Total LEDD mg: 307 (96.3)			
Cerasa et al. (2014)	12 ICD+ (11)	58 (8.3)	52.2 (7.6)		8.8 (5–13)	26.4 (10.3)	2 (1–3)	Total LEDD mg: 318.8* (150–800); DA agonist dosage: 1.1 (0.8)		MMSE, FAB, MWCST, Stroop test, Digit Span Forward and Backward, RAVLT, Token test, verbal fluency, BJLOT.	Yes/DSM IV
	12 ICD– (11)	59.3 (7.8)	54.8 (6.7)		12.1 (5–13)	25.9 (8.4)	2 (1–2)	Total LEDD mg: 300* 200–750); DA agonist dosage: 0.9 (0.7)			
Cilia et al. (2008)	11 ICD+ (10)	57.4 (5.8)	49.5 (4.7)	8.4 (3.4)	NR	18 (11)	2.1 (0.6)	Total LEDD mg: 811.8 (229); DA agonist dosage: 289.1 (57.5)	SOGS, diagnostic criteria,	MMSE, FAB, RCPM	Yes/MMSE
	40 ICD– (27)	55 (7)	46.4 (7.2)	8.4 (5.1)		19.1 (8.5)	2.3 (0.8)	Total LEDD mg: 877.3 (289.3); DA agonist dosage: 340.1 (157.2)			
Claassen et al. (2015)	12 ICD+ (8)	59.4 (5.5)	NR	6.5 (4.7)	17.1 (2.7)	15.9 (6.6)	NR	Total LEDD mg: 618.7 (361.9) Levodopa dosage mg/d: 408.2 (349.6) DA agonist dosage: 293.8 (167.4)	QUIP, diagnostic criteria	Stop signal task	Yes/MMSE
	12 ICD– (6)	60.8 (7.2)		6.1 (3.8)	16.3 (2.8)	15.7 (8.3)		Total LEDD mg: 520.3 (314.9) Levodopa dosage mg/d: 319.7 (318.9) DA agonist dosage: 200.6 (116.8)			
Djamshidian et al. (2010)	18 ICD+ (13)	55 (2.1)	43.9 (2.1)	10.9 (1.2)	12.2 (0.9)	18 (2.2)	NR	Total LEDD mg: 971 (183) Levodopa dosage mg/d: 752 (109)	diagnostic criteria	Digit span forward and backward, gambling task (see Huettel et al. 2006)	Yes/MMSE
	12 ICD– (9)	63.6 (2.2)	50.9 (2.2)	12.7 (2.1)	14.2 (1.3)	13 (1.4)		Total LEDD mg: 732 (203) Levodopa dosage mg/d: 604 (73)			

Djamshidian et al. (2011)	28 ICD+ (21) 24 ICD– (21)	54.6 (9.2) 64.2 (10.1)	44.5 (8.7) 52.5 (9.6)	10.1 (5.5) 11.7 (7.2)	13.4 (3) 14.7 (3.6)	15.5 (8.3) 14.4 (5.8)	NR	Total LEDD mg: 832 (425) Total LED mg: 821 (400)	diagnostic criteria	Stroop test	Yes/MMSE
Djamshidian et al. (2012)	26 ICD+ (22)	58.7 (10)	47.7 (9.5)	11.0 (4.1)	13.1 (2.8)	16.2 (10.6)	NR	Total LEDD mg: 934.2 (407)	diagnostic criteria, QUIP	Beads task, Working memory task,	Yes/MMSE
Housden et al. (2010)	27 ICD– (24) 18 ICD+ (11)	65.3 (5.3) 62.3 (7.6)	55.3 (7.4) NR	10.0 (6.5) 13.9 (9)	14.8 (2.5) NR	21.1 (9.0) 20 (6.6)	2.5 (0.6)	Total LEDD mg: 891.5 (432.1) Levodopa dosage mg/d: 643.5 (254.1) DA agonist dosage: 248 (301.3)	diagnostic criteria	Kirby delayed discounting questionnaire, Salience Attribution Test, MMSE, digit span forward and backward	Yes/MMSE
	18 ICD– (12)	67.7 (5.5)		12.9 (8.3)		21.3 (10.4)	2.5 (0.7)	Total LEDD mg: 804.8 (358.5) Levodopa dosage mg/d: 634.2 (301.7) DA agonist dosage: 170.5 (159.3)			
Hurt et al. (2014)	89 ICD+ (64)	63.2 (11.2)	NR	6* (0–25)	NR	25 (11.9)	2.4 (0.8)	Total LEDD mg: 880* (100–7565); Levodopa dosage mg/d: 520* (70–2660); DA agonist dosage: 330* (60–6340)	MIDI	ACE-R	Yes/ACE-R
	411 ICD– (262)	68.9 (9.9)		5* (0–39)		27.1 (12)	2.4 (0.9)	Total LEDD mg: 730* (37.5–3641); Levodopa dosage mg/d: 510* (50–2340); DA agonist dosage: 300* (25–2500)			
Leroi et al. (2011)	35 ICD+ 38 ICD–	NR	NR	NR	NR	26.85 (9.97) 24.11 (10.41)	NR	NR	diagnostic criteria	MMSE, <i>n</i> -back, phon-fluency	Yes/diagnostic criteria for PDD (Emre et al., 2007)
Leroi et al. (2013)	35 ICD+ (27)	58.9 (9.04)	56.68 (8.76)	Months:96.46 (52.23)	NR	26.85 (9.97)	2.21 (0.7)	Total LEDD mg: 994.68 (585.38) DA agonist dosage: 166.74 (175.23)	SOGS, diagnostic criteria	5-min word recall test, Phon- and semantic verbal fluency, WCST, TMT, serial 7's, digit <i>n</i> -back, National Adult Reading Test; Delay discounting task, Stop task.	NR
	55 ICD– (39)	62.54 (9.17)	54.25 (10.63)	97.56 (72.44)		28.93 (13.37)	2.40 (0.6)	Total LEDD mg: 732.25 (589.96) DA agonist dosage: 156.54 (161.11)			
Mack et al. (2013)	17 ICD+ (11)	61.1 (7.5)	48.1 (5.2)	13.1 (6.9)	NR	36.7 (16.1)	2.8 (1)	Total LEDD mg: 1677.9 (893.0)	diagnostic criteria	NART-R, MMSE, HVLT-R, Digit Span, Phon- and semantic verbal fluency, TMT	Yes/MMSE
	17 ICD– (8)	63.8 (8.5)	53.7 (10)	10.2 (5.6)		28.5 (15.2)	2.4 (1.3)	Total LEDD mg: 1269.3 (560.7)			
Pellicano et al. (2015)	18 ICD+ (16)	56.6 (9)	46.2 (10)	10.4 (4.8)	NR	OFF: 43 (11)	OFF 2.3 (0.5)	Total LEDD mg: 789 (370) Levodopa dosage mg/d: 644 (411.5) DA agonist dosage: 154 (144)	QUIP, diagnostic criteria	MMSE	Yes/the tool used to identify dementia was not specified
	18 ICD– (14)	55.8 (8)	49.4 (8)	6.4 (4.8)		28 (10)	1.9 (0.5)	Total LEDD mg: 560 (418) Levodopa dosage mg/d: 415 (397.3) DA agonist dosage: 124 (125)			
Pettorruo et al. (2014)	11 PG+ (8)	64.9 (10.9)	56.6 (10.6)	8.3 (3.2)	10 (4.2)	20.4 (12.3)	NR	Total LEDD mg: 712 (373) Levodopa dosage mg/d: 592 (404) DA agonist dosage: 120 (99)	diagnostic criteria	MMSE, FAB	Yes/diagnostic criteria for PDD (Emre et al., 2007)
	23 ICD+ (18)	62 (9.1)	53.2 (9)	8.8 (6)	11.3 (4.4)	18.4 (8.5)		Total LEDD mg: 654 (380) Levodopa dosage mg/d: 458 (376) DA agonist dosage: 196 (113)			
	120 ICD– (60)	67.7 (9.4)	60.6 (9.2)	7 (5.4)	11 (5.2)	20.4 (8.4)		Total LEDD mg: 575 (420) Levodopa dosage mg/d: 445 (386) DA agonist dosage: 130 (112)			
Pineau et al. (2016)	17 ICD+ (14)	55* (37–69)	48* (32–65)	7* (2–10)	7* (3–7)	7* (0–23)	NR	Total LEDD mg: 897.5* (299.88–1247.33) DA agonist dosage: 299.94* (77–718)	ASBPD	Conner's Performance Test, MCST, TMT, IGT	Yes/diagnostic criteria for PDD (Dubois et al., 2007)
	20 ICD– (13)	55* (40–62)	48* (35–55)	5.5* (4–12)	7* (3–7)	8.5* (0–34)		Total LEDD mg: 1049.89* (527.05–1549.84) DA agonist dosage: 340.23* (66.68–700)			

Table 2 (Continued)

Reference	Participants; (Gender: n Men)	mean age, years (SD)	mean age at PD onset, years (SD)	Mean PD duration (SD)	Mean Education (SD)	Mean UPDRS motor score	H&Y	Antiparkinsonian treatment	Tool assessing ICDs	Cognitive tasks	Dementia as exclusion criterion/tools used to identify dementia
Piray et al. (2014)	16 ICD+ (14)	64.38 (3.32)		9.63 (2.45)		19 (5.32)	2.47 (0.5)	NR	diagnostic criteria	MMSE, Digit span forward and backward, North American Adult Reading	Yes/MMSE
	15 ICD– (12) 25 ICD– (19)	63.33 (3.98) 63.92 (3.99)		8.87 (3.14) 9.72 (2.64)		19.6 (6.42) 20.36 (5.49)	2.40 (0.5) 2.5 (0.6)				
Politis et al. (2013)	12 HS+ (11)	55.2 (9.2)		9.6 (5.2)		23.1 (8.2)	NR	Total LEDD mg: 600 (327) Levodopa dosage mg/d: 288 (326) DA agonist dosage: 311 (183)	HS in according to criteria proposed by Voon et al.,	MMSE	Yes/the tool used to identify dementia was not specified
	12 HS+ (10)	62.3 (9.7)		10.1 (6.4)		20.0 (5.5)		Total LEDD mg: 778 (278) Levodopa dosage mg/d: 646 (264) DA agonist dosage: 132 (143)			
Pontieri et al. (2015)	21 PG+ (16)	58 (9)	51 (8)	8 (5)	10 (4)	21.5 (11.6)	2 (0.5)	Total LEDD mg: 794 (603) Levodopa dosage mg/d: 487 (625) DA agonist dosage: 307 (275)	QUIP, diagnostic criteria	MMSE, Phon- and semantic verbal fluency, RAVLT, WCST, Stroop Test, ROCF.	Yes/MMSE and diagnostic criteria for PDD (Emre et al., 2007)
	36 ICD+ (26)	64 (8)	57 (10)	7 (4)	11 (4)	19.1 (12.7)	1.9 (0.8)	Total LEDD mg: 704 (509) Levodopa dosage mg/d: 388 (278) DA agonist dosage: 316 (374)			
	98 ICD– (65)	66 (9)	61 (9)	5 (3)	10 (4)	19 (11.9)	1.8 (0.5)	Total LEDD mg: 416 (304) Levodopa dosage mg/d: 251 (279) DA agonist dosage: 166 (197)			
Rossi et al. (2010)	7 PG+ (6)	61.4 (6.9)	52 (5.6)	NR	13.8 (4.1)	17 (9.1)	2.2 (0.7)	Total LEDD mg: 935.9 (548.6) DA agonist dosage: 201.9 (78)	SOGS, diagnostic criteria	ACE-R, MMSE, FAB, WCST, Stroop, Social cognition, Go/no-go, Reversal and extension learning	Yes/MMSE
	13 PG– (10)	65.1 (3.8)	58.3 (6.9)		11.9 (5.5)	14.7 (6.7)	2 (0.7)	Total LEDD mg: 698.2 (4764.6) DA agonist dosage: 223.9 (136.8)			
Santangelo et al. (2009)	15 with PG (11)	61.87 (10.36)	53.2 (10.26)	8 (5.38)	9.53 (5.2)	15.07 (7.28)	1.93 (0.5)	Total LEDD mg: 774.17 (320.137) DA agonist dosage: 280 (209.42)	diagnostic criteria	MMSE, FAB, WCST, semantic and phonological fluency tasks, Corsi's test, Verbal Span, RCPM, ROCF, TMT; RAVLT	Yes/MMSE and DSM IV criteria
	15 PG– (11)	62.13 (8.87)	54.7 (9.4)	7.33 (4.40)	12.13 (4.8)	12.4 (6.18)	1.63 (0.6)	Total LEDD mg: 651 (284.05) DA agonist dosage: 292.66 (180.02)			
Siri et al. (2010)	21 ICD+ (18)	60.38 (7.6)	52.9 (8.6)	8.4 (4.5)	8.9 (4.1)	16.58 (7.7)	2.06 (0.7)	Total LEDD mg: 731.0 (283.9) Levodopa dosage mg/d: 503.9 (211.7) DA agonist dosage: 267.8 (113.5)	SOGS, diagnostic criteria	MMSE, FAB, RCPM, Phon and semantic fluency, RAVLT, digit span, corsi's test, Attentive matrices	no
	42 ICD– (24)	64.97 (5.9)	56.9 (7.3)	8.7 (3.9)	8.8 (4.6)	20.18 (12.5)	2.32 (0.3)	Total LEDD mg: 786.67 (283.9) Levodopa dosage mg/d: 598.5 (295.3) DA agonist dosage: 239.4 (130.5)			
Siri et al. (2015)	40 ICD+ (34)	59.2 (8.2)	50.1 (7.9)	9 (3.8)	9.6 (4)	16.7 (7.8)	NR	Total LEDD mg: 912.6 (297.0) Levodopa dosage mg/d: 695.5 (266.1) DA agonist dosage: 217.2 (144.1)	diagnostic criteria	MMSE, FAB, RCPM, Phon and semantic fluency, RAVLT, digit span forward and backward, corsi's test, attentive matrices	no
	40 ICD– (31)	62.3 (8.7)	53 (8.9)	9.3 (5)	10.5 (4.2)	17.6 (8.4)		Total LEDD mg: 760.2 (403.3) Levodopa dosage mg/d: 589.4 (429.2) DA agonist dosage: 170.7 (135.5)			
Tessitore et al. (2016)	15 ICD+ (13)	62.87 (8.6)	NR	5.3 (2.9)	9.8 (5)	10.9 (4.5)	1.3 (0.6)	Total LEDD mg: 477.3 (222.9) DA agonist dosage: 243.3 (82.1)	MIDI	MMSE, RAVLT, Corsi's test, ROCF, WCST, RCPM, Phon and semantic fluency, TMT, Stroop test, attentional matrices,	Yes/diagnostic criteria for PDD (Emre et al., 2007)
	15 ICD– (12)	63.14 (8)		6.6 (3.9)	12.9 (8)	12.1 (4.4)	1.4 (0.6)	Total LEDD: 477.3 (222.9) DA agonist dosage: 243.3 (82.1)			

Vitale et al. (2011)	13 HS+	68.7 (5.4)	59.5 (5.6)	8.5 (3.9)	9.5 (5)	15.1 (6.5)	1.8 (0.5)	Total LEDD mg: 727.3 (254.3) DA agonist dosage: 200 (130.4)	MIDI, diagnostic criteria	MMSE, WCST, ROCF, TMT, Attentive Matrices, Stroop, RAVLT.	Yes/MMSE and DSM IV criteria
	12 CE+	64.9 (7.5)	59.2 (7.1)	4.7 (2)	10.3 (5.8)	13 (6.9)	1.5 (0.5)	Total LEDD mg: 577.7 (306.2) DA agonist dosage: 250 (162)			
	14 PG+	62.8 (10.1)	54.4 (10)	7.6 (5.3)	9.9 (5.2)	15.4 (7.3)	2 (0.6)	Total LEDD mg: 761.6 (350.4) DA agonist dosage: 314.3 (288.5)			
	10 ICD+	62.2 (7.5)	55.5 (5.3)	8.1 (4.5)	8.2 (2.8)	13 (7.1)	1.5 (0.7)	Total LEDD mg: 808.3 (292.2) DA agonist dosage: 207.1 (159.2)			
	14 ICD–	61.3 (8.2)	53.2 (9.1)	7.6 (4.4)	13 (4)	11.7 (6)	1.8 (0.8)	Total LEDD mg: 630.3 (311.8) DA agonist dosage: 267.1 (201.3)			
Voon et al. (2007)	21 ICD+ (15)	60.2 (8.9)	50.9 (8.8)	9.2 (5.2)		15.2 (6.9)	2 (0.5)	Total LEDD mg 874.2 (495.6) DA agonist dosage: 268.3 (194.3)	diagnostic criteria	MMSE, FAB	Yes/DSM IV
	42 ICD– (21)	65.7 (9.9)	58.4 (10.1)	6.9 (4.2)		22.1 (13.9)	2.2 (0.8)	Total LEDD mg 746.9 (322.5) DA agonist dosage: 192.1 (105.3)			
Voon et al. (2010)	14 ICD+ (10)	51.52 (8.33)	NR	NR	12.02 (3.81)	NR	1.99 (0.5)	Total LEDD mg: 589.32 (301.25) DA agonist dosage: 161.53 (43.35)	diagnostic criteria	MMSE, Experiential discounting task, spatial working memory task and set-shifting task from the Cambridge Automated Neuropsychological Test Battery	Yes/DSM IV criteria
	14 ICD– (10)	54.51 (12.52)			11.12 (3.78)		2.35 (0.5)	Total LEDD mg: 609.55 (298.22) DA agonist dosage: 155.47 (57.35)			
Voon et al. (2011a)	14 ICD+ (10)	51.52 (8.33)	NR	NR	12.02 (3.81)	NR	1.99 (0.5)	Total LEDD mg: 589.32 (301.25) DA agonist dosage 161.53 (43.35)	diagnostic criteria	Risk task with a 'Gain' and 'Loss' condition	Yes/DSM IV criteria
	14 ICD– (10)	54.51 (12.52)			11.12 (3.78)		2.35 (0.5)	Total LEDD mg: 609.55 (298.22) DA agonist dosage 155.47 (57.35)			
Voon et al. (2011b)	282 ICD+ (191)	60.89 (0.50)	53.49 (0.59)	NR	NR	19.24 (0.74)	NR	Total LEDD mg: 946.1 (36.3) Levodopa dosage mg/d: 679.9 (33.6) DA agonist dosage: 266.3 (13.6)	diagnostic criteria	MoCA, delay discounting test	No
	282 ICD– (191)	61.34 (0.50)	53.90 (0.59)			19.62 (0.74)		Total LEDD mg: 809.2 (36.3) Levodopa dosage mg/d: 544.0 (33.6) DA agonist dosage: 265.2 (13.6)			
Yoo et al. (2015a)	10 punders (6)	66.8 (6.8)	NR	Months 84§ (42–144)	12*(1–16)	26.4 (10)	NR	Total LEDD mg: 495* (360–1000) DA agonist dosage: 160.0* (80–280)	diagnostic criteria	Seoul Neuropsychological Screening Battery, digit span forward and backward, Boston Naming test, ROCF, Go-nogo test, Stroop test, contrasting program, Phon and semantic fluency	Yes/diagnostic criteria for PDD (Emre et al., 2007)
	43 ICD– (25)	67.1 (6.6)		36§ (14–144)	9* (1–16)	26.8 (11.8)		Total LEDD mg: 450* (0–1540) DA agonist dosage: 150* (0–360)			
Yoo et al. (2015b)	10 ICD+ (7)	54.5 (6.2)	NR	10.2 (7.3)	NR	14.6 (11.5)	NR	Total LEDD mg: 924.6 (326.1) DA agonist dosage: 255.0 (177.6)	diagnostic criteria	MMSE	Yes/the tool used to identify dementia was not specified
	9 ICD– (6)	59.6 (8.6)		10.6 (3.9)		14.4 (8.0)		Total LEDD mg: 861.8 (440.6) DA agonist dosage: 170.0 (89.2)			

ICD, Impulse Control Disorders; SD, Standard Deviation; CE, Compulsive Eating; PG, Pathological Gambling; HS, Hypersexuality; M, Males; UPDRS, Unified Parkinson's Disease Rating Scale; Y&H, Hoehn & Yahr stage; MIDI, Minnesota Impulsive Disorders Interview; MMSE, Mini Mental State Examination; FAB, Frontal Assessment Battery, NR, Not Reported; RAVLT, Rey Auditory Verbal Learning Test; RCPM, Raven's Coloured Progressive Matrices; MWCST or MCST, Modified version of Wisconsin Card Sorting Test; Phon, Phonological fluency; IGT, Iowa Gambling Test; MoCA, Montreal Cognitive Assessment; BJLOT, Benton Judgement Orientation Lines Test, ACE-R, Addenbrookes Cognitive Examination-Revised version; TMT, Trail Making Test; ROCF, Rey–Osterrieth complex figure test; VOSP, Visual Object and Space Perception Battery; CDT, Clock Drawing Test, MCST, SOGS, South Oaks Gambling Screen, QUIP, Questionnaire for Impulsive–Compulsive Disorders in Parkinson's Disease; ASBPD, Arduin Scale of Behavior in Parkinson's Disease. * median; § months; ** OFF; NS, Not specified.

Table 3
Summary of meta-analytic results.

Domains/Outcomes	K	N	ICD+	ICD–	Pooled effect size Hedges'g (P value)	(95% Confidence Intervals)		Homogeneity statistics			Egger's t test for publication bias	Trim and fill
						LL	UL	Q (df)	P	I ²		
Abstraction ability*	8	459	212	247	–0.40 (0.002)	–0.65	–0.15	10.82 (7)	0.146	35.33	–0.58	0
EF- Fluid Reasoning	7	345	152	193	–0.23 (0.136)	–0.52	0.07	10.58 (6)	0.102	43.31	–3.64 (p = 0.015)	0
EF- Generativity-Fluency for letters	14	870	382	488	0.01 (0.937)	–0.23	0.25	35.92 (13)	0.001	63.81	–0.87	0
EF- Generativity-Fluency for categories	10	646	283	363	–0.08 (0.638)	–0.08	–0.24	32.56 (9)	<0.001	72.36	–1.49	0
EF-Inhibition	13	758	357	401	–0.10 (0.448)	–0.36	0.16	32.38 (12)	0.001	62.94	–0.5	0
EF-Set-shifting (TMT:B)	5	259	137	122	–0.59 (<0.001)	–0.84	–0.34	1.30 (4)	0.861	0	1	0
EF-Set-shifting (TMT:B-A)	7	421	224	197	–0.46 (0.043)	–0.091	–0.01	27.86 (6)	<0.001	78.47	–0.45	0
EF- Updating	7	309	146	163	0.26 (0.034)	0.02	0.5	6.53 (6)	0.367	8.09	0.38	0
EF-global	11	679	229	450	–0.08 (0.615)	–0.38	0.22	30.46 (10)	0.001	67.17	–0.02	2
												(–0.19 [–0.49–0.10])
Language	6	341	151	190	–0.24 (0.066)	–0.49	0.02	6.25 (5)	0.283	19.99	0.59	0
Processing speed, Attention, Working Memory	17	890	433	457	–0.10 (0.406)	–0.32	0.16	42.88 (16)	<0.001	62.69	–2.31 (p = 0.035)	0
Spatial Long term Memory	5	405	173	232	–0.33 (0.071)	–0.68	0.03	10.55 (4)	0.032	62.07	–1.4	0
Verbal Long Term Memory	12	732	343	389	–0.07 (0.633)	–0.38	0.23	40.15 (11)	<0.001	72.6	–1.26	1
												(–0.12 [–0.42–0.18])
Verbal Short Term Memory	10	514	240	274	–0.15 (0.324)	–0.45	0.15	23.91 (9)	0.004	62.36	0.37	1
												(–0.23 [–0.53–0.08])
Visuospatial/Constructional abilities	9	556	266	290	–0.42 (0.005)	–0.71	–0.13	19.32 (8)	0.013	58.59	–1.84	0
Decision making	10	352	171	181	0.54 (0.001)	0.23	0.85	18.56 (9)	0.029	51.51	1.31	0
Spatial short term memory	7	404	199	205	–0.10 (0.511)	–0.41	0.2	13.44 (6)	0.037	55.35	–0.98	0
Global Cognition	28	2492	960	1532	0.02 (0.781)	–0.11	0.15	49.48 (27)	0.005	45.43	–2.49 (p = 0.020)	0

*Abstraction ability/concept formation; EF, Executive Function; TMT, Trail Making Test; K = number of studies; N = total number of participants; ICD+ = number of PD patients with ICD; ICD– = number of PD patients without ICD; LL = Lower Limit; UL = Upper Limit; Q and I² = heterogeneity statistics; df = degrees of freedom; EF = Executive Functions; statistically significant values are reported in bold.

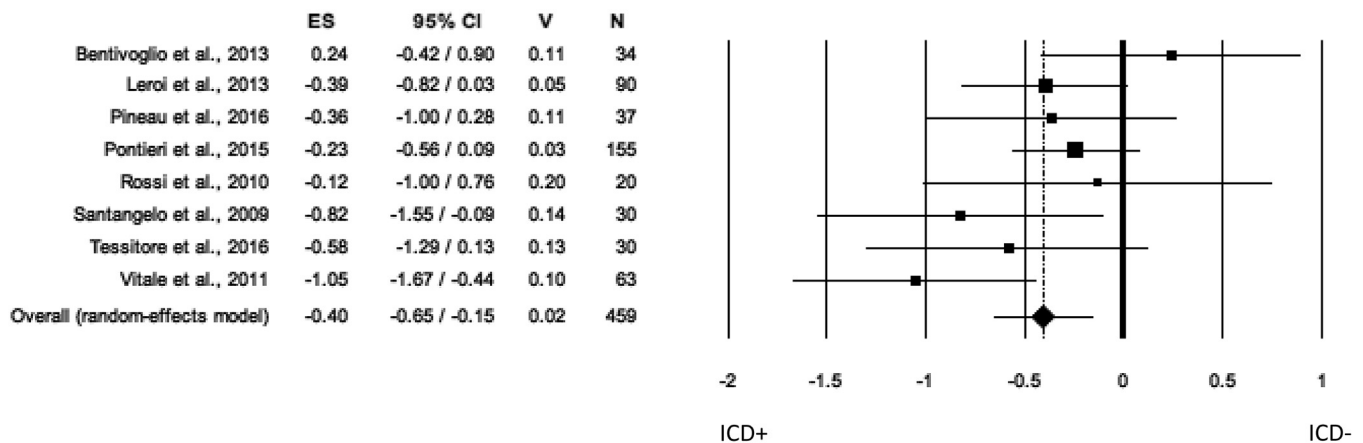


Fig. 2. Forest plot for abstraction/concept formation displaying effect size (Hedges' *g*) calculated using a random effects model.

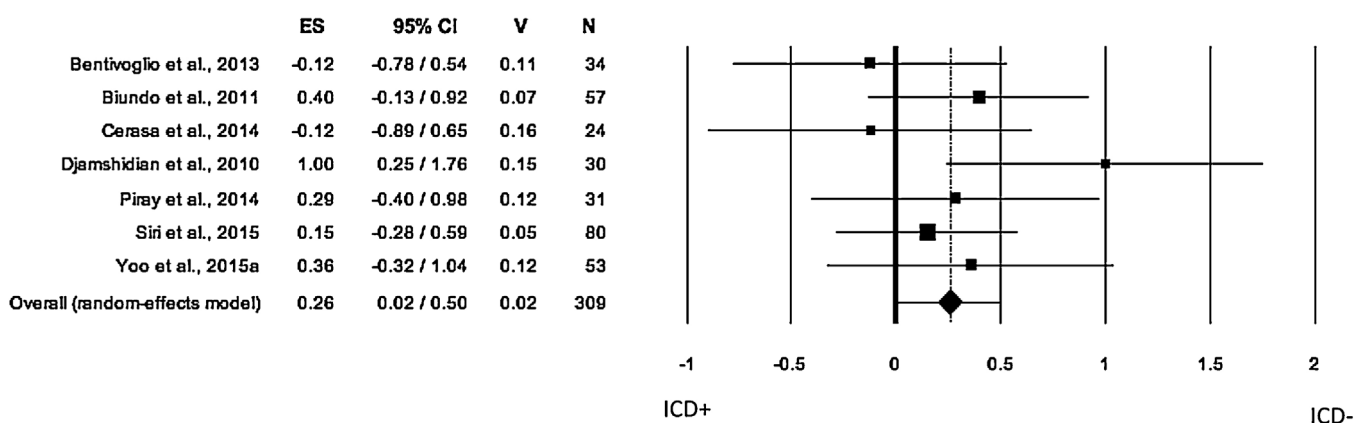


Fig. 3. Forest plot for updating displaying effect size (Hedges' *g*) calculated using a random effects model.

3.3. Moderator analysis

The moderation effects of demographic and clinical aspects of PD sample on outcome were evaluated by conducting meta-regression. According to Borenstein et al. (2009), we conducted meta-regression for outcomes in which there are at least 10 samples to 1 covariate: generativity (evaluated by semantic and phonological fluency), Inhibition, General Executive and Cognitive Functioning, Processing Speed/attention/working memory, Verbal Long Term memory, Verbal short term memory, and decision-making. Results from moderator analysis are shown in supplemental material 2. Meta-regression revealed that age at evaluation ($B = 0.17$, $p = 0.004$; $N = 10$ studies) had significant impact on the obtained effect size for verbal short-term memory; moreover, education was found to have significant impact on the obtained effect size for updating ($B = 0.26$, $p = 0.017$, $N = 6$ studies). Other demographic and clinical aspects did not show any significant effect on the effect size for this outcome. Finally, no other parameters had significant impact on the effect size for remaining outcomes.

4. Discussion

The present meta-analysis examined systematically the studies exploring cognitive profile of people affected by PD with ICD compared to those without ICD. It revealed that when cognition was measured by means of brief and global batteries (i.e. Mini Mental State Examination and Addenbrooke's Cognitive Evaluation) no significant association of ICD with either global cognitive ability or global frontal functioning was evident. On the other hand, when

we took into account specific cognitive functions, we found that people with PD and ICD showed more reduced abstraction ability/concept formation, set-shifting and visuospatial/constructional abilities compared with those without ICD. Moreover, people with PD and ICD made more impulsive and irrational choices than those without ICD.

Our finding of no significant association of ICD with either global cognitive ability or global frontal functioning (assessed by means of Frontal Assessment Battery, FAB alone) seems to be in line with results from previous studies. However, it is noteworthy that in 2 out of 11 studies (included in our meta-analysis) pathological gamblers with PD showed significantly poor performance on FAB than people with PD without ICD, suggesting that frontal dysfunctions can be a contributor for development and maintenance of PG (Santangelo et al., 2009; Pettorruso et al., 2014). The main difference between the above-mentioned two studies and others (9 studies) consisted in the composition of PG group: PG group enrolled by Santangelo et al. (2009) and Pettorruso et al. (2014) consisted of PD patients with PG alone, whereas other studies included many patients with PG associated with other ICD (i.e. HS, CE).

Taking into account specific cognitive domains, we found that ICD was significantly and moderately related to dysfunction of both set-shifting (i.e. ability to rapidly modify behavioural or cognitive dispositions in response to changing goals, to flexibly adapt to changing contexts) and decision making abilities. Although small, a relationship between ICD and reduced information updating ability (Updating) was also found in our meta-analysis. Set-shifting, Updating and Self-regulation, which allows to achieve long-term goals even if this requires delaying rewards (Goschke, 2014),

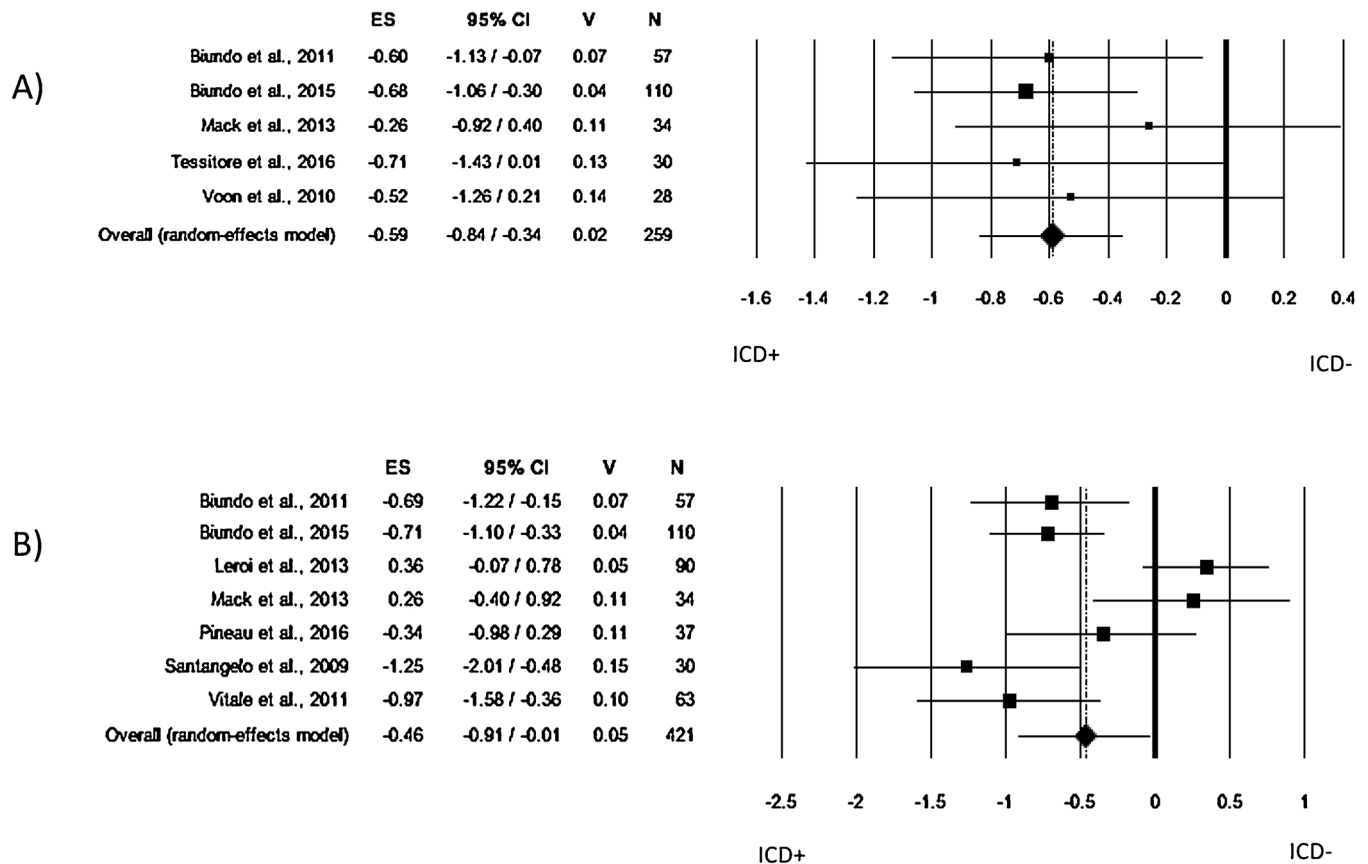


Fig. 4. Forest plot for set-shifting evaluated by Trail Making Test: part B (A) and part B-A (B) displaying effect size (Hedges' *g*) calculated using a random effects model.

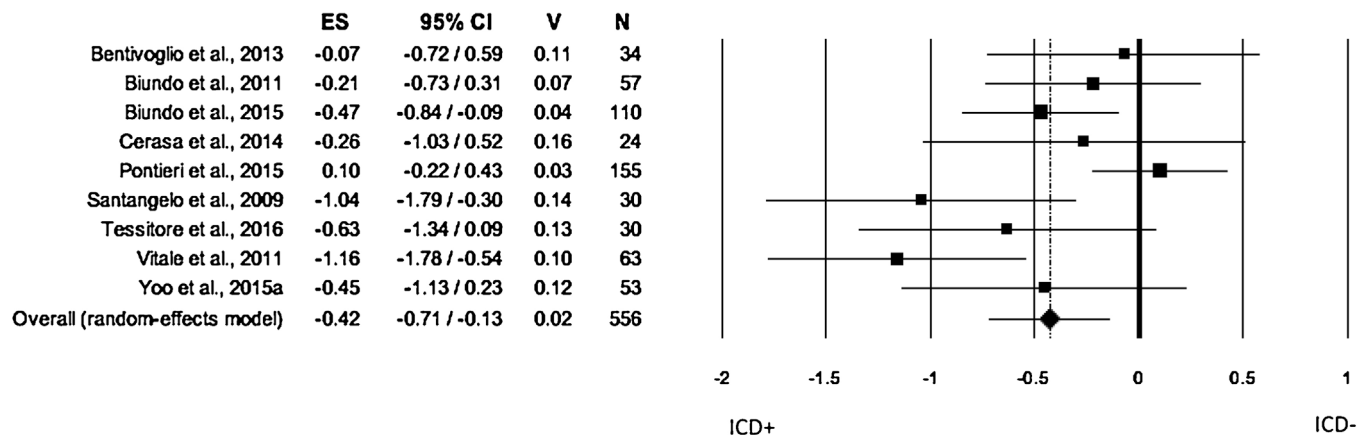


Fig. 5. Forest plot for visuospatial abilities displaying effect size (Hedges' *g*) calculated using a random effects model.

are executive functions playing a crucial role in selecting a goal and form intentions, maintaining actively goal representations; in modulating the (top-down) perceptual and response systems; in inhibiting habitual or impulsive responses; in rapidly updating and to flexibly switch both goals and behavioural dispositions; in controlling errors and conflicts; in regulation of the emotions (Goschke, 2014). Until now, behavioural studies have showed that pathological gamblers and patients with substance use dependence repeatedly ignore long-term negative consequences in order to obtain immediate gratification or relief from uncomfortable states associated with their addiction (Yechiam et al., 2005; van Holst et al., 2010a). In addition to disadvantageous decision-making, pathological gamblers have been described with less efficient

executive functions evidenced by working memory problems and impaired cognitive flexibility (see review of van Holst et al., 2010b). Studies in PD and non-PD samples revealed that subjects with a gambling disorder had higher score on anhedonia and apathy scale than those without PG (for relationship between anhedonia and ICD in PD and in non-PD: Pettorruso et al., 2014 and Thomsen et al., 2009, respectively; for relationship between apathy and ICD in PD: Leroi et al., 2012); on this basis gambling in anhedonic or apathetic PD patients was speculated to be a compulsive behavioural process, which aims at self-medicating the anhedonic state (Houeto et al., 2016). However, the findings of anhedonia and apathy in Parkinsonians with PG might be a further support to the idea that persistence of an impulsive and maladaptive behaviour in people

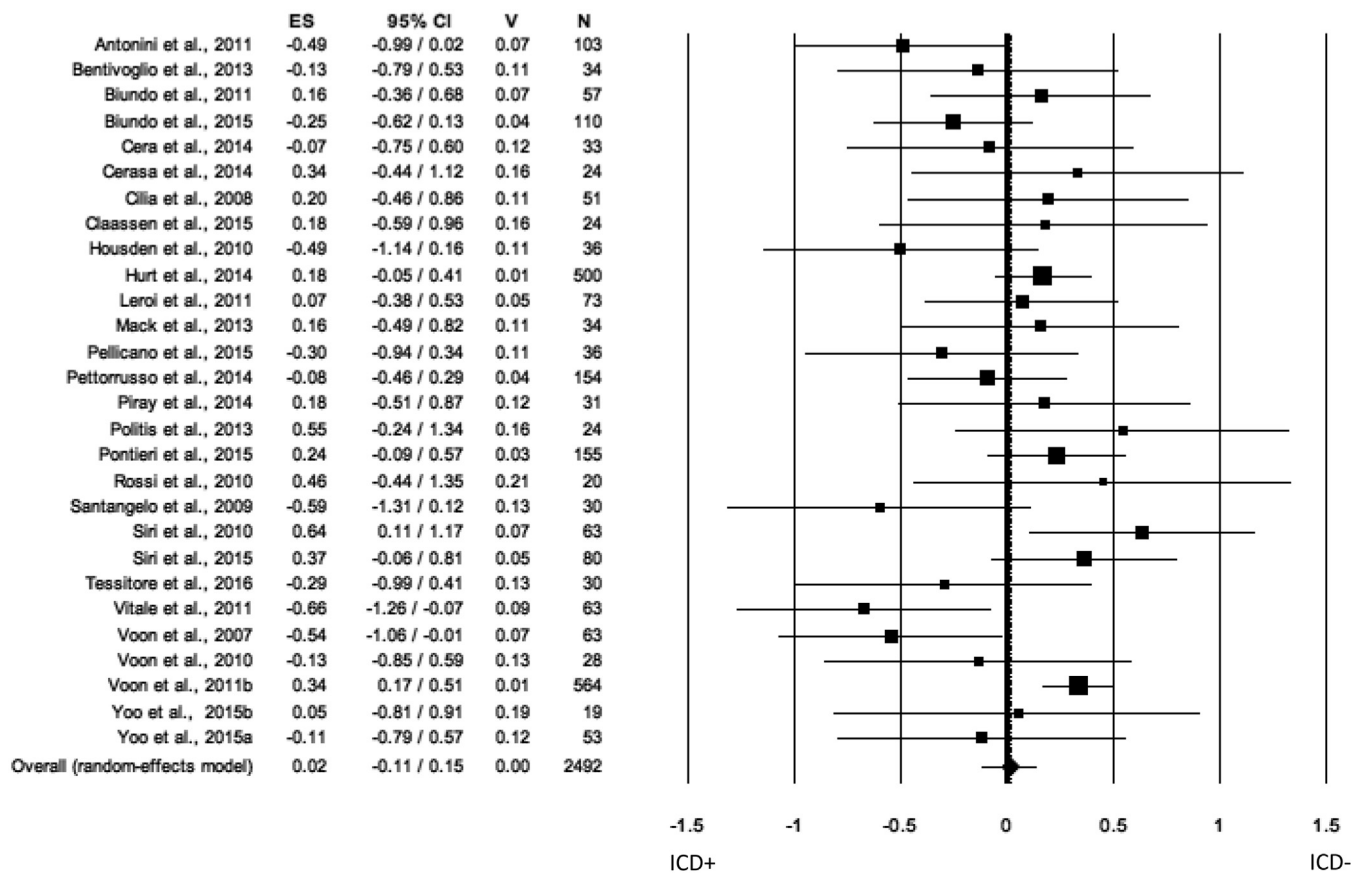


Fig. 6. Forest plot for global cognitive functioning displaying effect size (Hedges' g) calculated using a random effects model.

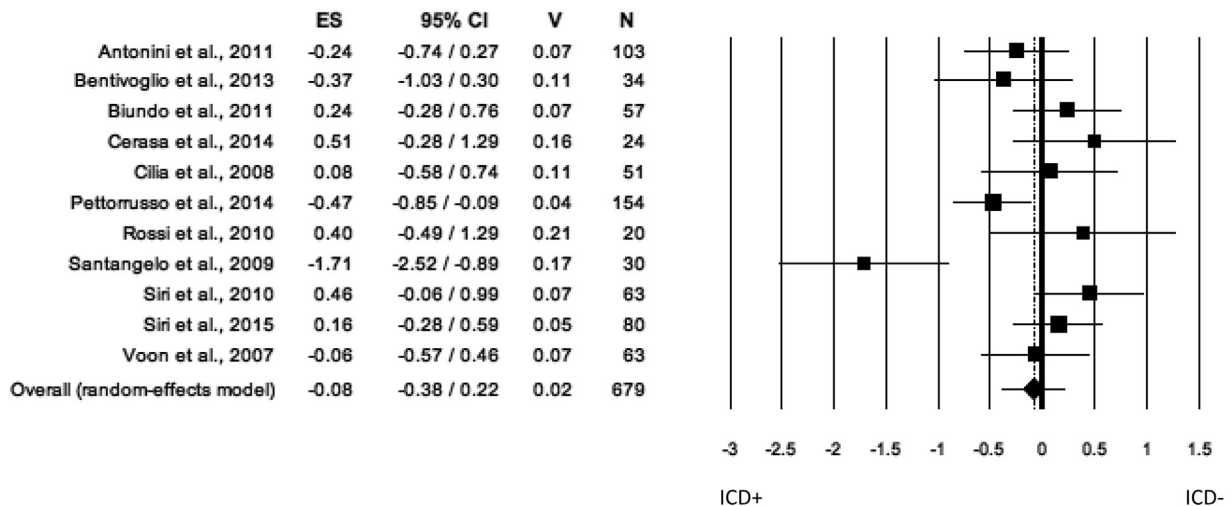


Fig. 7. Forest plot for frontal cognitive functioning displaying effect size (Hedges' g) calculated using a random effects model.

with PD depends on dysfunction of set-switching and cognitive rigidity as found in our meta-analysis and neuropsychological studies (Santangelo et al., 2009; Vitale et al., 2011; Pettorosso et al., 2014) rather than on rewarding nature of activities (i.e. gambling, eating).

The present meta-analytic review revealed a significant but small relationship between ICD and dysfunctional visuospatial/constructional abilities in PD. However, most of the studies included in the meta-analysis (7 out of 9) evaluated visuospatial/constructional functions by means of copy task of the Rey

Osterrieth Complex Figure, which is considered to be a useful tool for the evaluation of frontal lobe functions, such as strategic planning and organizing (Shin et al., 2006). Therefore, our findings could be interpreted as an additional support for the relationship between ICD and inefficient frontal functions.

The results of the present meta-analysis suggested the relevance of performing an early evaluation of specific cognitive functions to identify dysfunctions of cognitive flexibility and of self-regulation in subjects affected by PD who are susceptible to develop ICD or in subjects with active ICD. An early identification of cognitive dys-

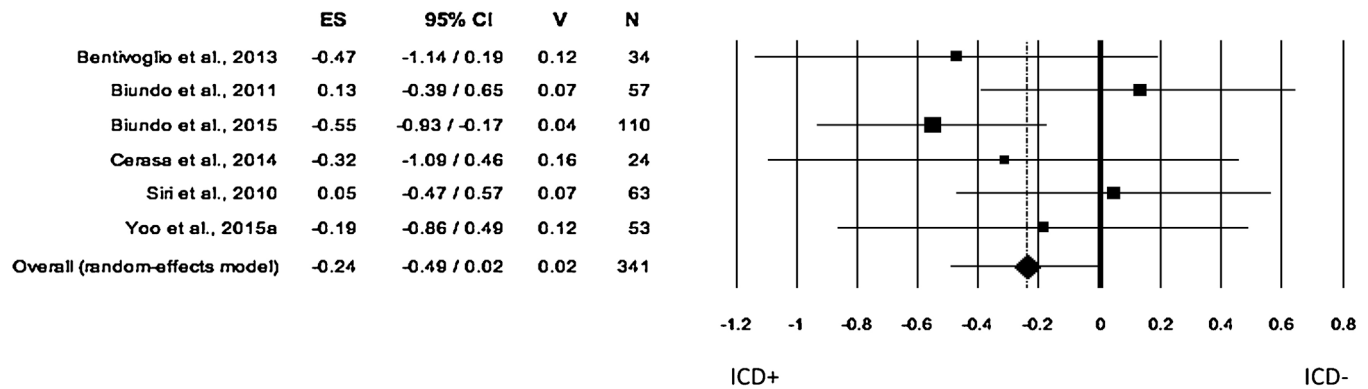


Fig. 8. Forest plot for language displaying effect size (Hedges' g) calculated using a random effects model.

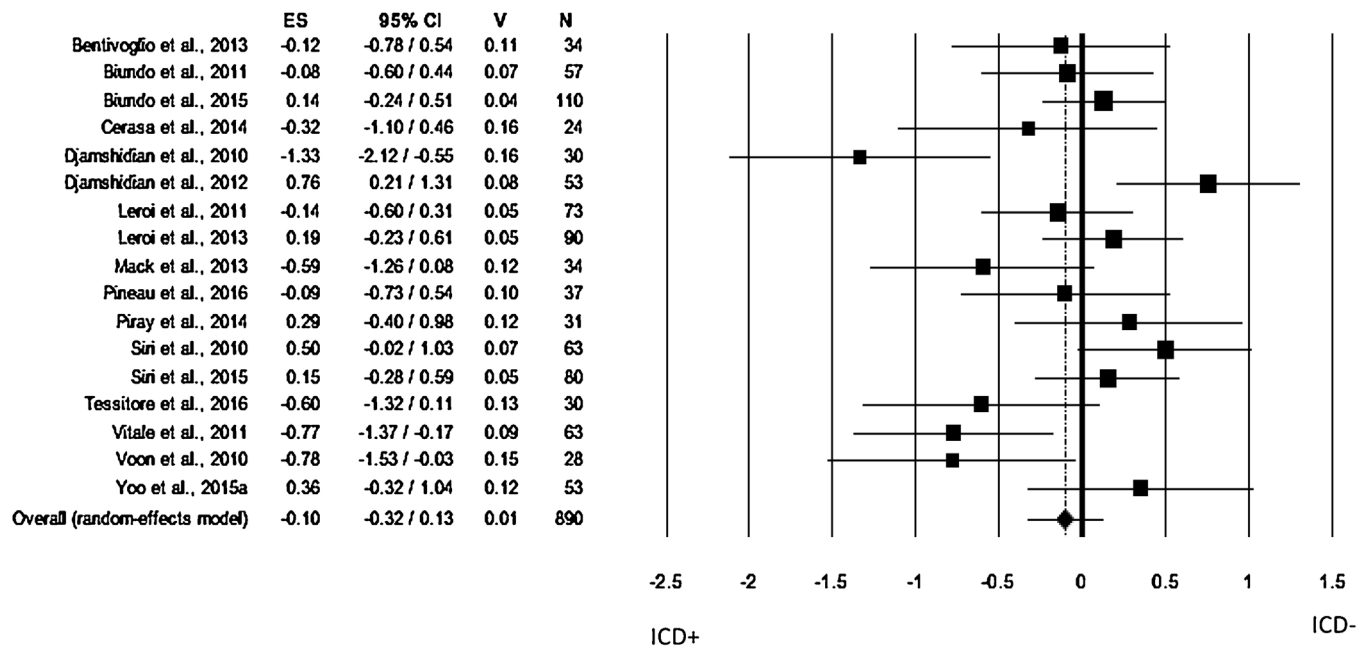


Fig. 9. Forest plot for processing speed/attention and working memory displaying effect size (Hedges' g) calculated using a random effects model.

functions could help clinicians to address a subject with PD and ICD to follow both a cognitive training, (i.e. a type of intervention which focuses upon improving cognitive function) and a psychological intervention such as cognitive-behavioural therapy (CBT). The CBT was reported to be clinically effective in the treatment of ICD in PD leading to a significant improvement in global ICB symptom severity at least in the short term (Okai et al., 2013). Instead, no study has explored the efficacy of a cognitive training on ICD in PD. However, since cognitive training has been reported to be efficacious on cognition in subjects with PD and to lead to an improvement of working memory, processing speed and executive functions (see meta-analysis of Leung et al., 2015), it is likely that a CT focused on set-shifting, decision making could be efficient to reduce the severity of ICDs even in people with PD.

Taken together, our results seem to be indirectly in line with previous neuroimaging evidences about association between ICD and functional abnormalities in cortical and subcortical regions involved in 'top-down' cognitive monitoring and inhibition of inappropriate behaviours (for a neuroimaging review see Ray and Strafella, 2013; Aracil-Bolaños and Strafella, 2016). Abnormalities in cortical and subcortical regions in PD are due to neurodegenerative processes originating from dorsal to ventral segments of striatum; therefore, whereas in early stage of disease loss

of nerve terminals occurs in the dorsal and posterior putamen, ventral regions are less affected (Vaillancourt et al., 2013). Exogenous dopamine administration could over-stimulate spared ventral regions triggering strong impulsive behaviours that at first give an immediate reward or pleasure (i.e. gambling, eating, shopping, or an increase in sexual thoughts, feelings and behaviours) despite significant impairment in social and occupational functioning. As dorsal striatal circuits are severely compromised, these impulsive and rewarding behaviours become compulsive and repetitive behaviours (habits) over time because of difficulty in monitoring, controlling, updating and modifying maladaptive behaviours, which can be a consequence of altered functioning of dorsal and ventral prefrontal-subcortical circuits (van den Heuvel et al., 2010).

Numerous studies have evidenced a strong association between ICD and antiparkinsonian drug use, male gender, young age at onset of PD, all factors found to increase risk for developing ICD (Weintraub et al., 2015). Therefore, these clinical aspects could influence the relationship between ICD and cognitive dysfunctions, too. Our results from moderator analysis did not reveal any effect of either demographic or clinical aspects on relationship between ICD and generativity (evaluated by semantic and phonological fluency), Inhibition, General Executive and Cognitive Functioning, Processing Speed/attention/working memory, Verbal Long Term memory,

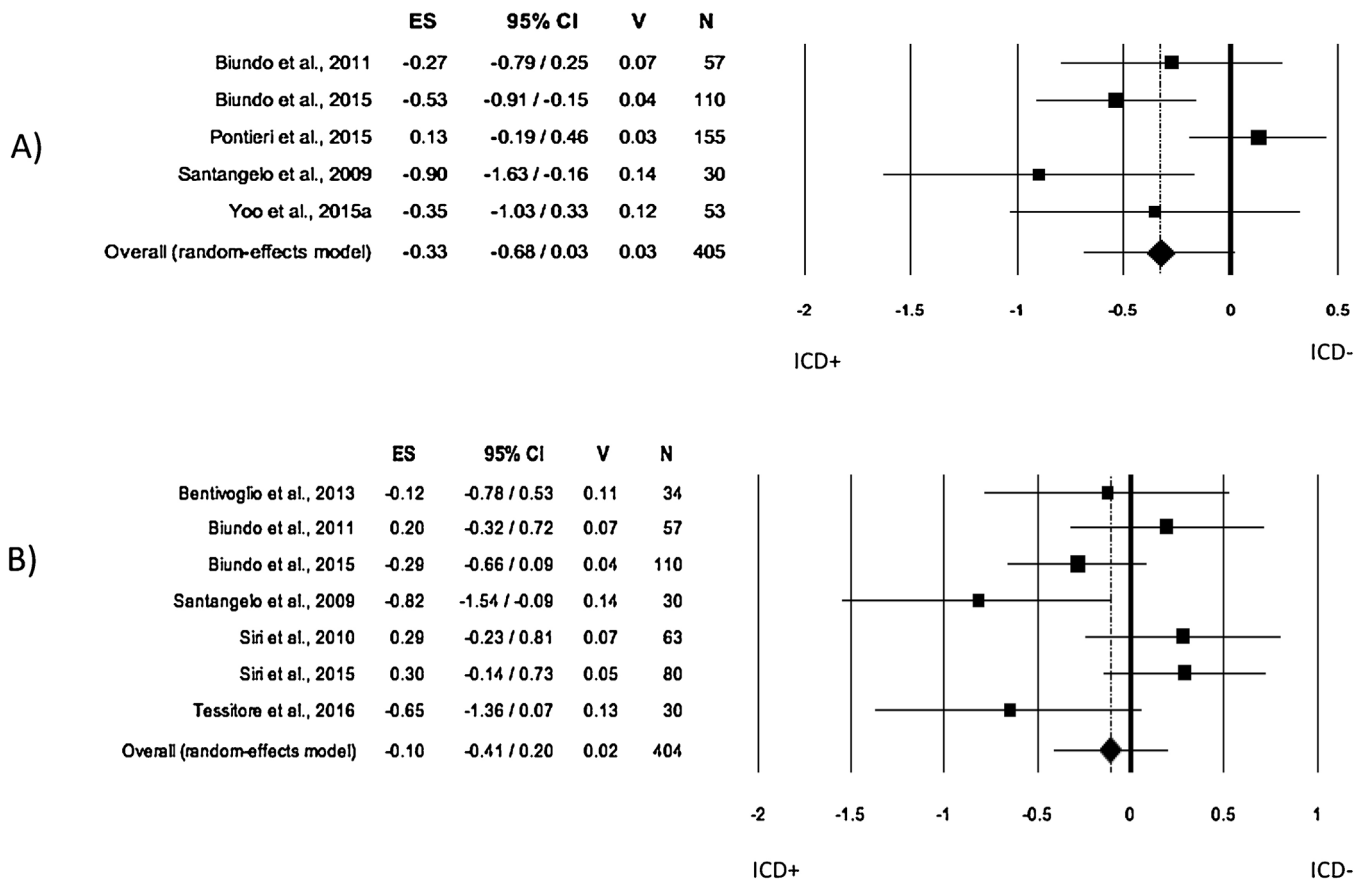


Fig. 10. Forest plot for memory evaluated by spatial long-term (A) and short-term (B) memory displaying effect size (Hedges' g) calculated using a random effects model.

and decision-making in PD, supporting the hypothesis of a specific neuropsychological pattern of people with PD affected by ICD, independent of demographic and clinical aspects. However, since clinical data of PD sample are not reported consistently across studies, these latter results should be interpreted with more caution; thus, the impact of these variables on relationship between ICD and cognition remains to be examined more deeply.

Our meta-analysis is characterized by some strengths: most studies included in the present meta-analytic review seem to be homogeneous since they evaluated occurrence of ICD using diagnostic clinical criteria, and that excluded patients with severe cognitive decline/dementia assessed by either clinical criteria or/and global cognitive battery (in detail, in 30/36 studies (83%) dementia was specified to be an exclusion criterion). Moreover, in each meta-analysis, sensitivity analyses indicated stability of meta-analytic findings. Finally, results from publication bias analyses conducted with the Egger's test and the trim and fill approach converged in supporting that the current data accurately represent the extant literature about neuropsychological profile associated with ICD in PD. However, some limitations need to be considered when examining the results of our meta-analytic review. The first is the choice of evaluating cognitive domains; this approach might be less informative than individual cognitive tasks, because cognitive tasks potentially evaluate different neuropsychological constructs. However, we followed guidelines for evaluation of cognitive domains proposed and published by Litvan et al. (2012). The second limit is that all studies included in the meta-analysis were cross-sectional studies; therefore, it was not possible to advance any causal inference about associations between cognitive deficit and ICD. Until now, only one follow-up study (Siri et al., 2015) compared the evolution of cognitive profile in PD patients with

and without ICD and found that ICD was not related to greater cognitive impairment or executive dysfunction, but rather to relatively lower cognitive decline over time. However, this study used neuropsychological tests which may have been insensitive to ICD-related deficits in PD and evaluated cognitive functions (i.e. fluid reasoning, general cognitive and frontal functioning, generativity, memory, attention), which did not seem to be impaired in ICD patients as revealed by findings from current meta-analytic review. Therefore, further longitudinal studies exploring evolution of set-switching, abstraction/concept formation, decision making processes are needed even to examine whether impairment of specific frontal functions could be predictors of development of ICD in people with PD.

Another limit of the present meta-analysis might be the pooling of the different ICD sub-types; until now, only one study (Vitale et al., 2011) compared the cognitive profile of different ICD subtypes (HS, CE, PG), and moreover, several studies included many patients with PG associated with other ICD (i.e. HS, CE). Therefore, because of limited number of studies including "pure PG" or "pure HS" or other pure type of ICD, here we could not perform a meta-analysis to evaluate the relationship between pure type of ICD and cognitive dysfunctions. Since cognitive and behavioural differences between subgroups were described in few previous studies (cognitive differences: Vitale et al., 2011; behavioural differences: Voon et al., 2011b) and may reflect differences between primary (eg, food, sex) and secondary (eg, money) rewards, degree of social interaction, or arousal associated with these behaviours (Voon et al., 2011b), further studies comparing cognitive and behavioural profiles of pure subtype of ICD are needed.

In conclusion, our findings support the hypothesis that ICDs occur in people with PD with specific frontal dysfunctions (i.e.

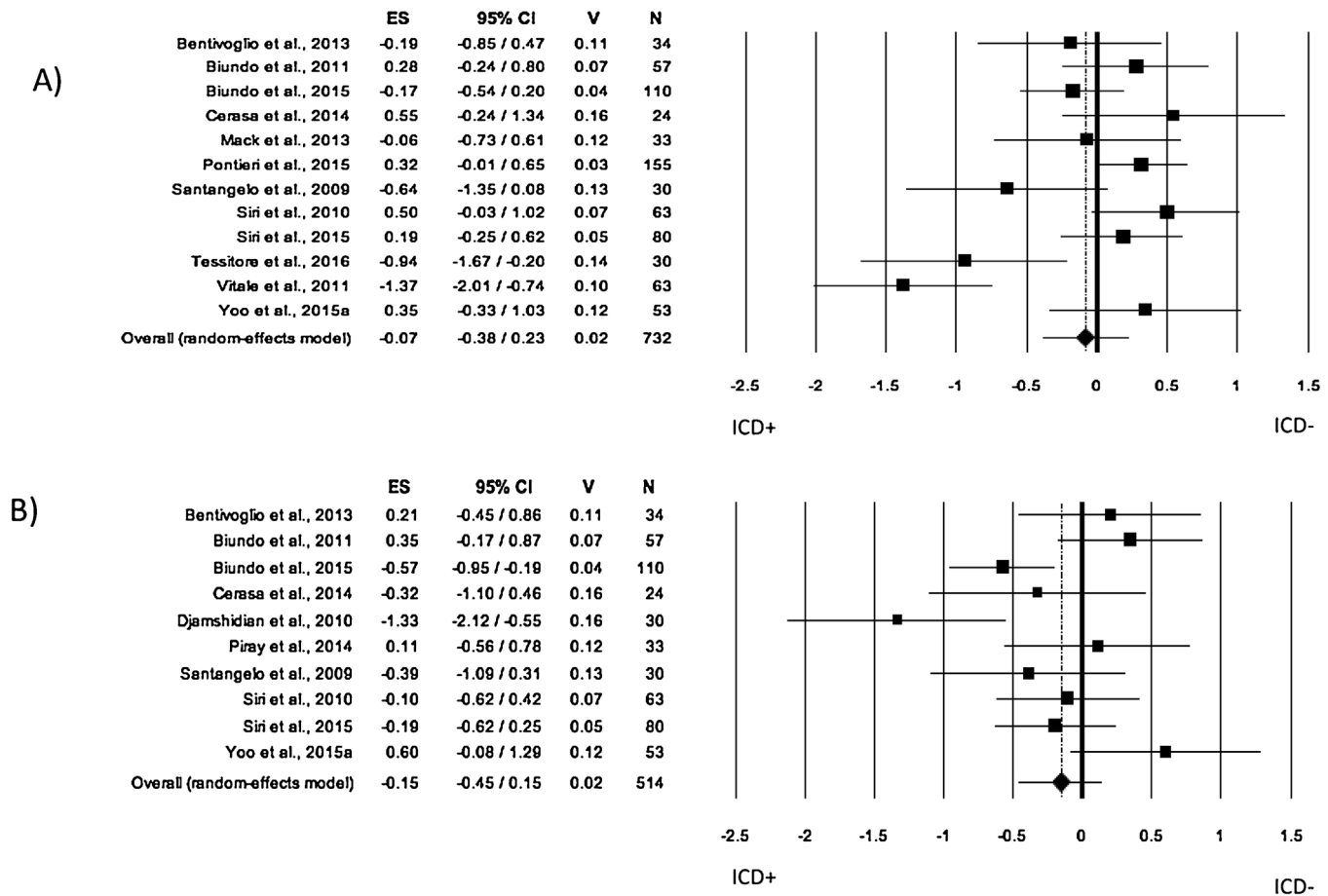


Fig. 11. Forest plot for memory evaluated by verbal long-term (A) and short-term (B) memory displaying effect size (Hedges' g) calculated using a random effects model.

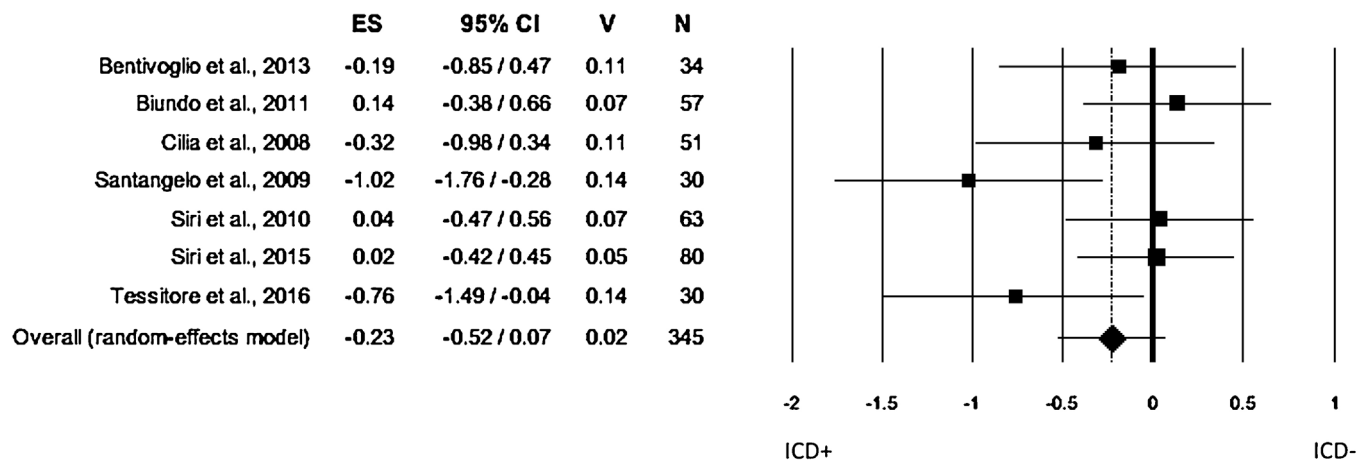


Fig. 12. Forest plot for fluid reasoning displaying effect size (Hedges' g) calculated using a random effects model.

altered set-switching, concept formation, decision making) taking antiparkinsonian drug use. These results are compatible with the idea that DA treatment can trigger ICD in susceptible people with PD with defect of set-switching, concept formation, decision making due to an imbalance of prefrontal-subcortical limbic circuitries (Napier et al., 2015).

Thus, in the clinical practice it may be useful to assess different cognitive functions such as set-switching, abstraction/concept formation and decision making before treating people with PD with antiparkinsonian drugs to detect those individuals more susceptible to an ICD when taking dopaminergic drugs.

Author role

Santangelo Gabriella: 1. Research project: A. Conception, B. Organization, C. Execution; 2. Statistical Analysis: A. Design, B. Execution, C. Review and Critique; 3. Manuscript Preparation: A. Writing of the first draft, B. Review and Critique;

Raimo Simona: 1. Research project: C. Execution; 2. C. Review and Critique; 3. B. Review and Critique

Paolo Barone: 1. Research project: A. Conception, B. Organization; 2. Statistical Analysis: C. Review and Critique; 3. Manuscript Preparation: B. Review and Critique.

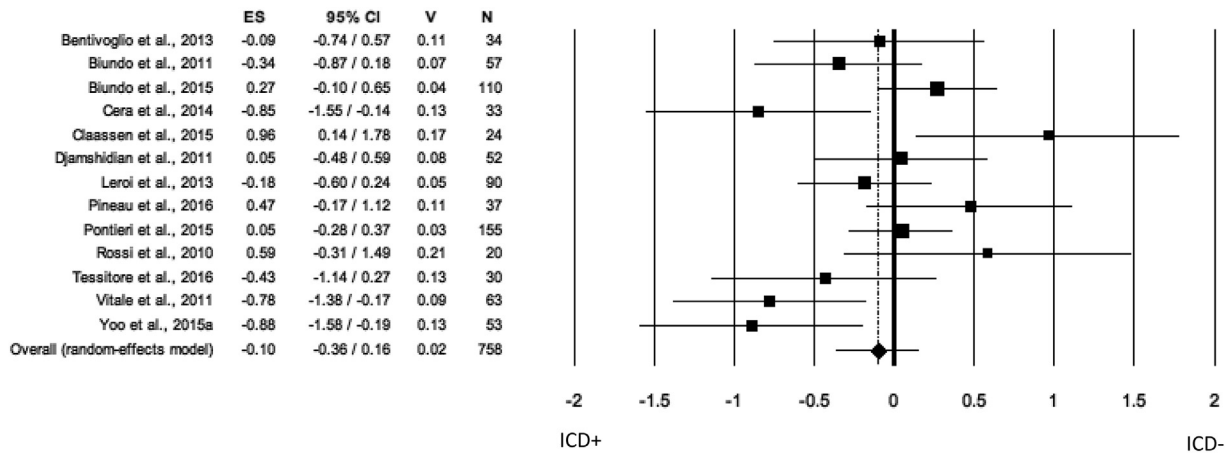


Fig. 13. Forest plot for inhibition displaying effect size (Hedges' g) calculated using a random effects model.

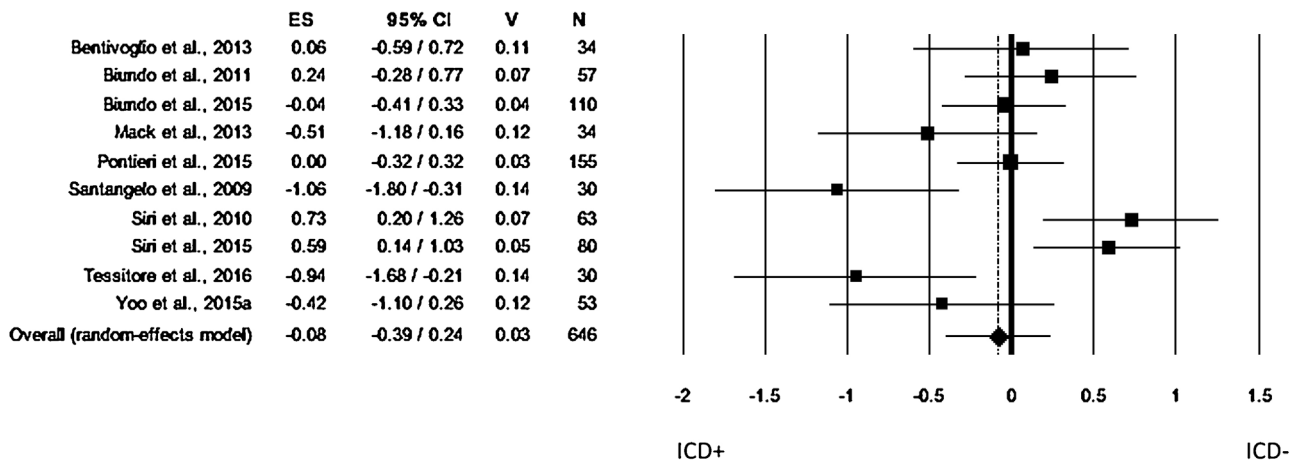


Fig. 14. Forest plot for generativity (evaluated by semantic fluency task) displaying effect size (Hedges' g) calculated using a random effects model.

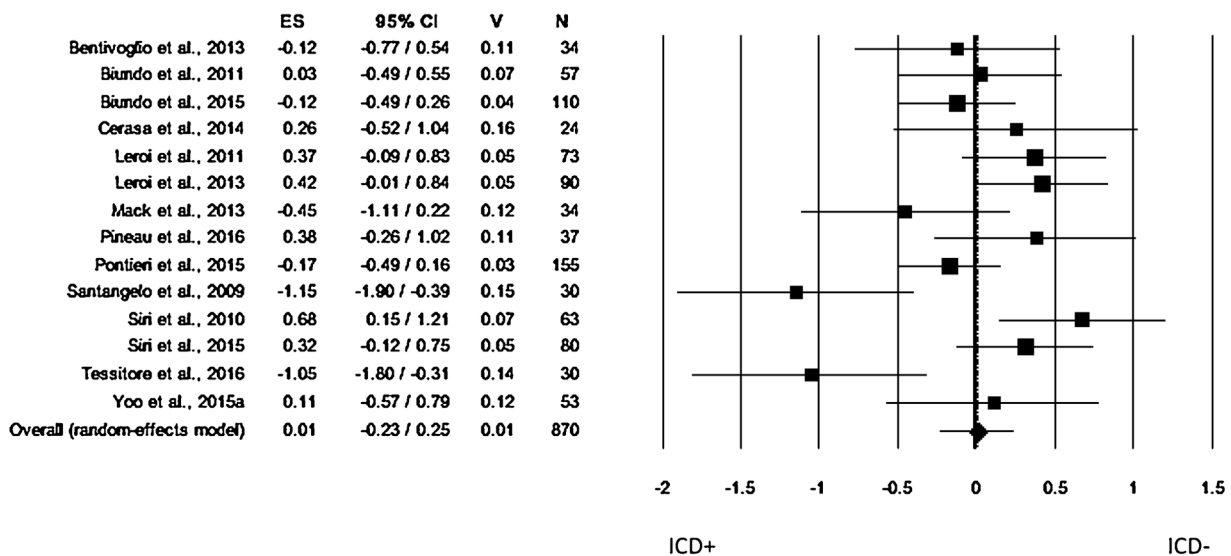


Fig. 15. Forest plot for generativity (evaluated by phonological fluency task) displaying effect size (Hedges' g) calculated using a random effects model.

	ES	95% CI	V	N
Bentivoglio et al., 2013	0.37	-0.29 / 1.03	0.11	34
Cera et al., 2014	0.13	-0.54 / 0.81	0.12	33
Djamshidian et al., 2010	0.25	-0.47 / 0.96	0.13	30
Djamshidian et al., 2012	0.82	0.27 / 1.37	0.08	53
Housden et al., 2010	0.78	0.12 / 1.44	0.11	36
Leroi et al., 2013	0.47	-0.08 / 1.02	0.08	53
Pineau et al., 2016	0.12	-0.52 / 0.75	0.10	37
Rossi et al., 2010	1.82	0.78 / 2.87	0.28	20
Voon et al., 2010	-0.12	-0.84 / 0.60	0.13	28
Voon et al., 2011a	1.36	0.56 / 2.16	0.17	28
Overall (random-effects model)	0.54	0.23 / 0.85	0.02	352

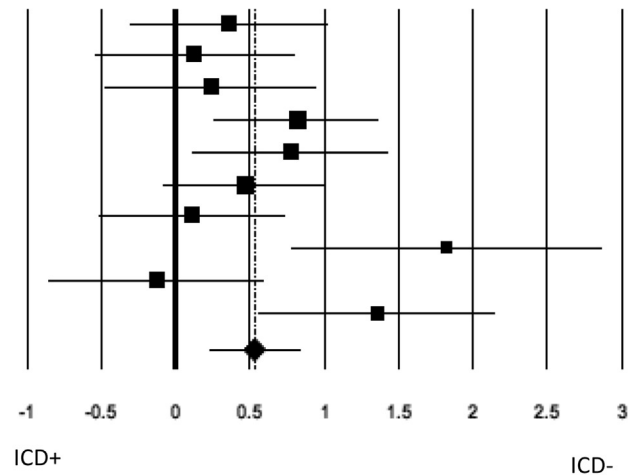


Fig. 16. Forest plot for decision making displaying effect size (Hedges' g) calculated using a random effects model.

Full financial disclosure for the previous 12 months

Gabriella Santangelo receives a salary from University of Campania Luigi Vanvitelli. Simona Raimo stated no financial disclosure. Paolo Barone has served as a consultant for Chiesi, Lundbeck, UCB Pharma, and Zambon; he receives a salary from University of Salerno; and receives a grant from the Michael J Fox Foundation.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.neubiorev.2017.02.018>.

References

- Antonini, A., Siri, C., Santangelo, G., Cilia, R., Poletti, M., Canesi, M., Caporali, A., Mancini, F., Pezzoli, G., Ceravolo, R., Bonuccelli, U., Barone, P., 2011. *Impulsivity and compulsivity in drug-naïve patients with Parkinson's disease*. *Mov. Disord.* 26 (3), 464–468.
- Aracil-Bolaños, I., Strafella, A.P., 2016. *Molecular imaging and neural networks in impulse control disorders in Parkinson's disease*. *Parkinsonism Relat. Disord.* 22 (Suppl. 1), S101–S105.
- Bentivoglio, A.R., Baldonero, E., Ricciardi, L., De Nigris, F., Daniele, A., 2013. *Neuropsychological features of patients with Parkinson's disease and impulse control disorders*. *Neurol. Sci.* 34 (7), 1207–1213.
- Biundo, R., Formento-Dojot, P., Facchini, S., Vallerluna, A., Ghezzi, L., Foscolo, L., Meneghello, F., Antonini, A., 2011. *Brain volume changes in Parkinson's disease and their relationship with cognitive and behavioural abnormalities*. *J. Neurol. Sci.* 310 (1–2), 64–69.
- Biundo, R., Weis, L., Facchini, S., Formento-Dojot, P., Vallerluna, A., Pilleri, M., Weintraub, D., Antonini, A., 2015. *Patterns of cortical thickness associated with impulse control disorders in Parkinson's disease*. *Mov. Disord.* 30 (5), 688–695.
- Borenstein, M., Hedges, L.V., Higgins, J.P., Rothstein, H.R., 2009. *Introduction to Meta-analysis*. Wiley, Chichester.
- Cera, N., Bifulchetti, S., Martinotti, G., Gambi, F., Sepede, G., Onofri, M., Di Giannantonio, M., Thomas, A., 2014. *Amantadine and cognitive flexibility: decision making in Parkinson's patients with severe pathological gambling and other impulse control disorders*. *Neuropsychiatr. Dis. Treat.* 10, 1093–1101.
- Cerasa, A., Salsone, M., Nigro, S., Chiriac, C., Donzuso, G., Bosco, D., Vasta, R., Quattrone, A., 2014. *Cortical volume and folding abnormalities in Parkinson's disease patients with pathological gambling*. *Parkinsonism Relat. Disord.* 20 (11), 1209–1214.
- Cilia, R., Siri, C., Marotta, G., Isaia, I.U., De Gaspari, D., Canesi, M., Pezzoli, G., Antonini, A., 2008. *Functional abnormalities underlying pathological gambling in Parkinson disease*. *Arch. Neurol.* 65 (12), 1604–1611.
- Claassen, D.O., van den Wildenberg, W.P., Harrison, M.B., van Wouwe, N.C., Kanoff, K., Neimat, J.S., Wylie, S.A., 2015. *Proficient motor impulse control in Parkinson disease patients with impulsive and compulsive behaviors*. *Pharmacol. Biochem. Behav.* 129, 19–25.
- Cohen, J., 1988. *Statistical Power Analysis for the Behavioral Sciences*. Academic Press, New York.
- Djamshidian, A., Jha, A., O'sullivan, S.S., Silveira-Moriyama, L., Jacobson, C., Brown, P., Lees, A., Averbeck, B.B., 2010. *Risk and learning in impulsive and nonimpulsive patients with Parkinson's disease*. *Mov. Disord.* 25 (13), 2203–2210.
- Djamshidian, A., O'sullivan, S.S., Lees, A., Averbeck, B.B., 2011. *Stroop test performance in impulsive and non impulsive patients with Parkinson's disease*. *Parkinsonism Relat. Disord.* 17 (3), 212–214.
- Djamshidian, A., O'sullivan, S.S., Sanotsky, Y., Sharman, S., Matviyenko, Y., Foltynie, T., Michalczuk, R., Aviles-Olmos, I., Fedoryshyn, L., Doherty, K.M., Filts, Y., Selikhova, M., Bowden-Jones, H., Joyce, E., Lees, A.J., Averbeck, B.B., 2012. *Decision making, impulsivity, and addictions: do Parkinson's disease patients jump to conclusions?* *Mov. Disord.* 27 (9), 1137–1145.
- Dubois, B., Burn, D., Goetz, C., Aarsland, D., Brown, R.G., Broe, G.A., Dickson, D., Duyckaerts, C., Cummings, J., Gauthier, S., Korczyn, A., Lees, A., Levy, R., Litvan, I., Mizuno, Y., McKeith, I.G., Olanow, C.W., Poewe, W., Sampaio, C., Tolosa, E., Emre, M., 2007. *Diagnostic procedures for Parkinson's disease dementia: recommendations from the movement disorder society task force*. *Mov. Disord.* 22 (16), 2314–2324.
- Duval, S., 2005. *The trim and fill method*. In: Rothstein, H.R., Sutton, A.J., Borenstein, M. (Eds.), *Publication Bias in Meta-analysis*. Wiley, Chichester, pp. 11–33.
- Duval, S., Tweedie, R., 2000. *A nonparametric 'trim and fill' method of accounting for publication bias in meta-analysis*. *J. Am. Stat. Assoc.* 95, 89–98.
- Egger, M., Davey Smith, G., Schneider, M., Minder, C., 1997. *Bias in meta-analysis detected by a simple, graphical test*. *BMJ* 315 (7109), 629–634.
- Emre, M., Aarsland, D., Brown, R., Burn, D.J., Duyckaerts, C., Mizuno, Y., Broe, G.A., Cummings, J., Dickson, D.W., Gauthier, S., Goldman, J., Goetz, C., Korczyn, A., Lees, A., Levy, R., Litvan, I., McKeith, I., Olanow, W., Poewe, W., Quinn, N., Sampaio, C., Tolosa, E., Dubois, B., 2007. *Clinical diagnostic criteria for dementia associated with Parkinson's disease*. *Mov. Disord.* 22 (12), 1689–1707.
- Goschke, T., 2014. *Dysfunctions of decision-making and cognitive control as transdiagnostic mechanisms of mental disorders: advances, gaps, and needs in current research*. *Int. J. Methods. Psychiatr. Res.* 23 (Suppl. 1), 41–57.
- Higgins, J.P., Thompson, S.G., Deeks, J.J., Altman, D.G., 2003. *Measuring inconsistency in meta-analysis*. *BMJ* 327 (7414), 557–560.
- Houeto, J.L., Magnard, R., Dalley, J.W., Belin, D., Carnicella, S., 2016. *Trait impulsivity and anhedonia: two gateways for the development of impulse control disorders in parkinson's disease?* *Front. Psychiatry* 7, 91.
- Housden, C.R., O'sullivan, S.S., Joyce, E.M., Lees, A.J., Roiser, J.P., 2010. *Intact reward learning but elevated delay discounting in Parkinson's disease patients with impulsive-compulsive spectrum behaviors*. *Neuropsychopharmacology* 35 (11), 2155–2164.
- Hozo, S.P., Djulbegovic, B., Hozo, I., 2005. *Estimating the mean and variance from the median, range, and the size of a sample*. *BMC. Med. Res. Methodol.* 5, 13.
- Huedo-Medina, T.B., Sánchez-Meca, J., Marín-Martínez, F., Botella, J., 2006. *Assessing heterogeneity in meta-analysis: Q statistic or I2 index?* *Psychol. Methods* 11 (2), 193–206.
- Huettel, S.A., Stowe, C.J., Gordon, E.M., Warner, B.T., Platt, M.L., 2006. *Neural signatures of economic preferences for risk and ambiguity*. *Neuron* 49 (5), 765–775.
- Hurt, C.S., Alkufri, F., Brown, R.G., Burn, D.J., Hindle, J.V., Landau, S., Wilson, K.C., Samuel, M., PROMS-PD study group, 2014. *Motor phenotypes, medication and mood: further associations with impulsive behaviours in Parkinson's disease*. *J. Parkinsons Dis.* 4 (2), 245–254.
- Leeman, R.F., Potenza, M.N., 2012. *Similarities and differences between pathological gambling and substance use disorders: a focus on impulsivity and compulsivity*. *Psychopharmacology (Berl.)* 219, 469–490.

- Leroi, I., Ahearn, D.J., Andrews, M., McDonald, K.R., Byrne, E.J., Burns, A., 2011. Behavioural disorders, disability and quality of life in Parkinson's disease. *Age Ageing* 40 (5), 614–621.
- Leroi, I., Andrews, M., McDonald, K., Harbisetar, V., Elliott, R., Byrne, E.J., Burns, A., 2012. Apathy and impulse control disorders in Parkinson's disease: a direct comparison. *Parkinsonism Relat. Disord.* 18 (2), 198–203.
- Leroi, I., Barraclough, M., McKie, S., Hinvest, N., Evans, J., Elliott, R., McDonald, K., 2013. Dopaminergic influences on executive function and impulsive behaviour in impulse control disorders in Parkinson's disease. *J. Neuropsychol.* 7 (2), 306–325.
- Leung, I.H., Walton, C.C., Hallock, H., Lewis, S.J., Valenzuela, M., Lampit, A., 2015. Cognitive training in Parkinson's disease: a systematic review and meta-analysis. *Neurology* 85 (21), 1843–1851.
- Litvan, I., Goldman, J.G., Tröster, A.I., Schmand, B.A., Weintraub, D., Petersen, R.C., Mollenhauer, B., Adler, C.H., Marder, K., Williams-Gray, C.H., Aarsland, D., Kulisevsky, J., Rodriguez-Oroz, M.C., Burn, D.J., Barker, R.A., Emre, M., 2012. Diagnostic criteria for mild cognitive impairment in Parkinson's disease: movement Disorder Society Task Force guidelines. *Mov. Disord.* 27 (3), 349–356.
- Mack, J., Okai, D., Brown, R.G., Askey-Jones, S., Chaudhuri, K.R., Martin, A., Samuel, M., David, A.S., 2013. The role of self-awareness and cognitive dysfunction in Parkinson's disease with and without impulse-control disorder. *J. Neuropsychiatry. Clin. Neurosci.* 25 (2), 141–149.
- Napier, T.C., Corvol, J.C., Grace, A.A., Roitman, J.D., Rowe, J., Voon, V., Strafella, A.P., 2015. Linking neuroscience with modern concepts of impulse control disorders in Parkinson's disease. *Mov. Disord.* 30 (2), 14114–14119, <http://dx.doi.org/10.1002/mds.26068>, Review. PubMed PMID: 25476402; PubMed Central PMCID: PMC4318759.
- Okai, D., Askey-Jones, S., Samuel, M., O'sullivan, S.S., Chaudhuri, K.R., Martin, A., Mack, J., Brown, R.G., David, A.S., 2013. Trial of CBT for impulse control behaviors affecting Parkinson patients and their caregivers. *Neurology* 80 (9), 792–799.
- Pellicano, C., Niccolini, F., Wu, K., O'sullivan, S.S., Lawrence, A.D., Lees, A.J., Piccini, P., Politis, M., 2015. Morphometric changes in the reward system of Parkinson's disease patients with impulse control disorders. *J. Neurol.* 262 (12), 2653–2661.
- Pettorosso, M., Martinotti, G., Fasano, A., Loria, G., Di Nicola, M., De Risio, L., Ricciardi, L., Conte, G., Janiri, L., Bentivoglio, A.R., 2014. Anhedonia in Parkinson's disease patients with and without pathological gambling: a case-control study. *Psychiatry Res.* 215 (2), 448–452.
- Pineau, F., Roze, E., Lacomblez, L., Bonnet, A.M., Vidailhet, M., Czernecki, V., Corvol, J.C., 2016. Executive functioning and risk-taking behavior in Parkinson's disease patients with impulse control disorders. *J. Neural Transm. (Vienna)* 123 (6), 573–581.
- Piray, P., Zeighami, Y., Bahrami, F., Eissa, A.M., Hewedi, D.H., Moustafa, A.A., 2014. Impulse control disorders in Parkinson's disease are associated with dysfunction in stimulus valuation but not action valuation. *J. Neurosci.* 34 (23), 7814–7824.
- Politis, M., Loane, C., Wu, K., O'sullivan, S.S., Woodhead, Z., Kiferle, L., Lawrence, A.D., Lees, A.J., Piccini, P., 2013. Neural response to visual sexual cues in dopamine treatment-linked hypersexuality in Parkinson's disease. *Brain* 136 (Pt. 2), 400–411.
- Pontieri, F.E., Assogna, F., Pellicano, C., Cacciari, C., Pannunzi, S., Morrone, A., Danese, E., Caltagirone, C., Spalletta, G., 2015. Sociodemographic, neuropsychiatric and cognitive characteristics of pathological gambling and impulse control disorders NOS in Parkinson's disease. *Eur. Neuropsychopharmacol.* 25 (1), 69–76.
- Ray, N.J., Strafella, A.P., 2013. Imaging impulse control disorders in Parkinson's disease and their relationship to addiction. *J. Neural Transm. (Vienna)* 120 (4), 659–664.
- Rossi, M., Gerschovich, E.R., de Achaval, D., Perez-Lloret, S., Cerquetti, D., Cammarota, A., Inés Nouzeilles, M., Fahrer, R., Merello, M., Leiguarda, R., 2010. Decision-making in Parkinson's disease patients with and without pathological gambling. *Eur. J. Neurol.* 17 (1), 97–102.
- Rothstein, H.R., Sutton, A.J., Borenstein, M., 2005. *Publication Bias in Meta-analysis*. Wiley, Chichester.
- Santangelo, G., Vitale, C., Trojano, L., Verde, F., Grossi, D., Barone, P., 2009. Cognitive dysfunctions and pathological gambling in patients with Parkinson's disease. *Mov. Disord.* 24 (6), 899–905.
- Santangelo, G., Barone, P., Trojano, L., Vitale, C., 2013a. Pathological gambling in Parkinson's disease. A comprehensive review. *Parkinsonism Relat. Disord.* 19 (7), 645–653.
- Santangelo, G., Trojano, L., Barone, P., Grossi, D., Vitale, C., 2013b. Impulse control disorders and cognitive dysfunctions in patients with Parkinson's disease. *Neurol. Sci.* 34 (11), 2045–2046.
- Shin, M.S., Park, S.Y., Park, S.R., Seol, S.H., Kwon, J.S., 2006. Clinical and empirical applications of the Rey-Osterrieth complex figure test. *Nat. Protoc.* 1 (2), 892–899.
- Siri, C., Cilia, R., De Gaspari, D., Canesi, M., Meucci, N., Zecchinelli, A.L., Pezzoli, G., Antonini, A., 2010. Cognitive status of patients with Parkinson's disease and pathological gambling. *J. Neurol.* 257 (2), 247–252.
- Siri, C., Cilia, R., Reali, E., Pozzi, B., Cereda, E., Colombo, A., Meucci, N., Canesi, M., Zecchinelli, A.L., Tesi, S., Mariani, C.B., Sacilotto, G., Zini, M., Pezzoli, G., 2015. Long-term cognitive follow-up of Parkinson's disease patients with impulse control disorders. *Mov. Disord.* 30 (5), 696–704.
- Tessitore, A., Santangelo, G., De Micco, R., Vitale, C., Giordano, A., Raimo, S., Corbo, D., Amboni, M., Barone, P., Tedeschi, G., 2016. Cortical thickness changes in patients with Parkinson's disease and impulse control disorders. *Parkinsonism Relat. Disord.* 24, 119–125.
- Thomsen, K.R., Callesen, M.B., Linnet, J., Kringelbach, M.L., Moller, A., 2009. Severity of gambling is associated with severity of depressive symptoms in pathological gamblers. *Behav. Pharmacol.* 20 (5–6), 527–536.
- Vaillancourt, D.E., Schonfeld, D., Kwak, Y., Bohnen, N.I., Seidler, R., 2013. Dopamine overdose hypothesis: evidence and clinical implications. *Mov. Disord.* 28 (14), 1920–1929.
- Vitale, C., Santangelo, G., Trojano, L., Verde, F., Rocco, M., Grossi, D., Barone, P., 2011. Comparative neuropsychological profile of pathological gambling, hypersexuality, and compulsive eating in Parkinson's disease. *Mov. Disord.* 26 (5), 830–836.
- Voon, V., Thomsen, T., Miyasaki, J.M., de Souza, M., Shafro, A., Fox, S.H., Duff-Canning, S., Lang, A.E., Zurofski, M., 2007. Factors associated with dopaminergic drug-related pathological gambling in parkinson disease. *Arch. Neurol.* 64 (2), 212–216.
- Voon, V., Reynolds, B., Brezing, C., Gallea, C., Skaljic, M., Ekanayake, V., Fernandez, H., Potenza, M.N., Dolan, R.J., Hallett, M., 2010. Impulsive choice and response in dopamine agonist-related impulse control behaviors. *Psychopharmacology (Berl.)* 207, 645–659.
- Voon, V., Gao, J., Brezing, C., Symmonds, M., Ekanayake, V., Fernandez, H., Dolan, R.J., Hallett, M., 2011a. Dopamine agonists and risk: impulse control disorders in Parkinson's disease. *Brain* 134 (Pt. 5), 1438–1446.
- Voon, V., Sohr, M., Lang, A.E., Potenza, M.N., Siderowf, A.D., Whetteckey, J., Weintraub, D., Wunderlich, G.R., Stacy, M., 2011b. Impulse control disorders in Parkinson disease: a multicenter case-control study. *Ann. Neurol.* 69 (6), 986–996.
- Weintraub, D., David, A.S., Evans, A.H., Grant, J.E., Stacy, M., 2015. Clinical spectrum of impulse control disorders in Parkinson's disease. *Mov. Disord.* 30 (2), 121–127.
- Yechiam, E., Busemeyer, J.R., Stout, J.C., Bechara, A., 2005. Using cognitive models to map relations between neuropsychological disorders and human decision-making deficits. *Psychol. Sci.* 16 (12), 973–978.
- Yoo, H.S., Yun, H.J., Chung, S.J., Sunwoo, M.K., Lee, J.M., Sohn, Y.H., Lee, P.H., 2015a. Patterns of neuropsychological profile and cortical thinning in parkinson's disease with punning. *PLoS One* 10 (7), e0134468.
- Yoo, H.B., Lee, J.Y., Lee, J.S., Kang, H., Kim, Y.K., Song, I.C., Lee, D.S., Jeon, B.S., 2015b. Whole-brain diffusion-tensor changes in parkinsonian patients with impulse control disorders. *J. Clin. Neurol.* 11 (1), 42–47.
- van den Heuvel, O.A., van der Werf, Y.D., Verhoef, K.M., de Wit, S., Berendse, H.W., Wolters E.Ch. Veltman, D.J., Groenewegen, H.J., 2010. Frontal-striatal abnormalities underlying behaviours in the compulsive-impulsive spectrum. *J. Neurol. Sci.* 289 (1–2), 55–59.
- van Holst, R.J., van den Brink, W., Veltman, D.J., Goudriaan, A.E., 2010a. Why gamblers fail to win: a review of cognitive and neuroimaging findings in pathological gambling. *Neurosci. Biobehav. Rev.* 34 (1), 87–107.
- van Holst, R.J., van den Brink, W., Veltman, D.J., Goudriaan, A.E., 2010b. Brain imaging studies in pathological gambling. *Curr. Psychiatry Rep.* 12 (5), 418–425.